
Scalable Neural Safety Certification via Monotonicity

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Abstract

Learning-based safety certification offers a promising route for verifying complex dynamical systems when explicit models are unavailable, but existing data-driven methods often scale poorly: they treat the system as a black box, rely on dense discretizations or Lipschitz bounds, and consequently suffer from sample complexity that grows exponentially with dimension. This paper shows that a common structural property of dynamical systems—*monotonicity*—can be used to break this curse of dimensionality. We develop a data-driven framework for robust safety verification and safe controller synthesis for unknown monotone systems using only simulator queries. Our main theoretical result proves that, for monotone systems with upper-closed unsafe sets, safety is completely characterized by the existence of monotone inductive barrier certificates. This structure reduces certificate verification over continuous state spaces to localized boundary checks over finitely many cells. To learn such certificates, we introduce Max-Monotone Networks, a monotone neural architecture with universal approximation guarantees for monotone functions. We then propose a training and refinement algorithm that, upon successful termination, returns formally valid neural barrier certificates and, under mild partition assumptions, requires only $O(n)$ simulator queries. Across high-dimensional benchmarks, including oscillator networks with up to 13,659 states and traffic networks with 1,000 states, our method certifies safety in regimes where prior approaches either do not converge or exceed memory limits. These results demonstrate that exploiting order structure can make neural safety certification both scalable and formally sound.

1 Introduction

Dynamical systems provide a powerful mathematical framework for modeling and analyzing autonomous, engineered, and natural systems. They are widely used to describe complex behaviors arising in safety-critical domains such as self-driving vehicles, biological networks, and industrial process control. As autonomous systems become increasingly prevalent, it is essential to develop rigorous methods that can certify their safe operation over long or infinite time horizons. Barrier certificates provide a powerful inductive approach for this purpose: they reduce safety verification to the search for a scalar function that separates initial states from unsafe states and is preserved along system trajectories [42, 4]. When such a function exists, it serves as a formal certificate that the unsafe set is unreachable.

Although barrier certificates provide a clean inductive proof of safety, they remain difficult to compute for high-dimensional systems, especially in data-driven settings where the dynamics are unknown

and only simulator-generated data or trajectories are available. Classical barrier-certificate methods construct certificates using linear programming [46, 58], sum-of-squares (SOS) optimization [42], or satisfiability-modulo-theories (SMT) techniques [25]. These approaches have led to substantial progress and provide strong formal guarantees [41, 5, 55], but they often require explicit models, restrictive dynamics, or optimization procedures that scale poorly with dimension. Neural networks have recently emerged as a flexible representation class for learning barrier certificates from data [14, 63, 62, 45]. However, existing neural barrier frameworks typically rely on Lipschitz bounds, dense sampling, or expensive verification subroutines to obtain formal guarantees. As a result, their sample complexity and computational cost can grow rapidly with the state dimension.

This paper shows that a fundamental structural property of dynamical systems—*monotonicity*—can be used to overcome this scalability bottleneck. Monotonicity means that the system preserves a natural partial order on states and inputs throughout its evolution. This property has been widely studied in dynamical systems and formal methods as a way to capture cooperative behavior and monotone structure in complex systems [28, 12]. Monotone dynamics arise naturally in many real-world settings, including traffic networks [19, 17], queuing systems [27], population dynamics [31], chemical reaction networks [8], and biological systems [54]. In many of these settings, monotonicity follows *naturally* from the physical or network structure of the system. Moreover, monotonicity has also emerged as a *design principle* for ensuring closed-loop performance and robustness [35, 9]. Even when a system is not inherently monotone, controllers can often be designed to enforce monotonicity in the closed-loop dynamics, as in droop control for power systems [50], consensus protocols in multi-agent robotics [10], and structural control strategies in biological networks [9, 54].

A central advantage of monotone dynamics is that order preservation makes many global questions reducible to extremal cases over order intervals, a principle that underlies scalable analysis and control methods for this class of systems [52, 9]. These properties have been exploited to establish stability of interconnected systems [9], enable efficient control synthesis [44, 24], and support scalable reachability analysis [18]. Despite this rich body of work, the role of monotonicity in data-driven safety certification remains much less developed, with only limited recent progress on neural barrier certificates for monotone systems [2]. In particular, it has remained unclear whether monotonicity can provide a complete barrier-certificate characterization of safety, whether such certificates can be learned from simulator queries alone, and whether the resulting verification procedure can avoid the exponential sample complexity typical of black-box methods.

Our key observation is that monotonicity turns the certificate-learning problem from a global search over the state space into a finite set of boundary conditions. For monotone systems with upper-closed unsafe sets, we prove a complete characterization: safety holds if and only if there exists a *monotone* inductive barrier certificate. Moreover, the monotonicity of the certificate and dynamics allows global safety constraints over continuous state and input spaces to be reduced to localized boundary checks over finitely many cells. Thus, instead of densely sampling the full state space, safety certification can be performed using carefully selected simulator queries.

Building on this result, we develop a learning-based framework for robust safety verification and safe controller synthesis for unknown monotone systems. We introduce *Max-Monotone Networks*, a monotone neural architecture with universal approximation guarantees for monotone functions. The architecture is designed so that the learned neural certificate respects the order structure required by our theory. We then propose a training and refinement algorithm that uses simulator queries to learn neural barrier certificates satisfying the finite boundary conditions. Upon successful termination, the algorithm returns a formally valid certificate of safety. Under mild partition assumptions, the number of simulator queries required by the refinement procedure scales *linearly* in the system dimension.

Our contributions are summarized as follows:

- We prove that, for monotone discrete-time systems with upper-closed unsafe sets, robust safety verification and safe controller synthesis admit complete characterizations via *monotone* inductive barrier certificates.
- We show that monotonicity reduces global certificate verification over continuous state and input spaces to finite localized boundary checks, enabling formal safety certification from simulator queries.
- We introduce *Max-Monotone Networks*, a monotone neural architecture with universal approximation guarantees, tailored to learning monotone barrier certificates.

- We develop a training and adaptive refinement algorithm that returns formally valid neural barrier certificates upon successful termination and achieves $O(n)$ simulator-query complexity under mild partition assumptions.
- We demonstrate scalability on high-dimensional benchmarks, including traffic-flow models, Kuramoto oscillator networks, power-system models, Lotka–Volterra dynamics, and ribosome-flow models. Our method certifies systems with thousands of dimensions, including a 13,659-state oscillator network and a 1,000-state traffic network, while competing approaches either fail to converge or exceed memory limits.

Together, these results show that exploiting order structure can turn neural safety certification from a high-dimensional black-box sampling problem into a scalable, formally certified learning procedure.

2 Preliminaries and problem statement

In this section, we review the necessary preliminaries and formulate the main problem studied in this paper. We refer to Appendix B for a summary of the notation used throughout the paper.

2.1 Dynamical systems

Dynamical systems provide a fundamental framework for describing the evolution of natural or engineered processes over time.

Definition 2.1 (Dynamical systems). A *discrete-time dynamical system* (dtDS) is a tuple $\mathfrak{S} := (\mathcal{X}, \mathcal{X}_0, U, f)$, where $\mathcal{X} \subseteq \mathbb{R}^n$ denotes the set of states, $\mathcal{X}_0 \subseteq \mathcal{X}$ is the set of initial states, $U \subseteq \mathbb{R}^m$ is the set of inputs, and $f : \mathcal{X} \times U \rightarrow \mathcal{X}$ is the state-transition map.

For an input sequence $\{u_t\}_{t=0}^\infty \in U$, the evolution of the system $\mathfrak{S}$ is given by

$$x_{t+1} = f(x_t, u_t), \quad \forall t \geq 0. \quad (1)$$

The resulting state sequence $\{x_t\}_{t=0}^\infty$ is called the *trajectory* of $\mathfrak{S}$ associated with the input $\{u_t\}_{t=0}^\infty$.

Safety is a fundamental property of dynamical systems. Given a dynamical system $\mathfrak{S}$, we adopt a set-theoretic perspective on its safety by specifying a set $\mathcal{X}_u \subseteq \mathcal{X}$ of unsafe states. The system $\mathfrak{S}$ is safe if its state remains outside $\mathcal{X}_u$ for all time.

Definition 2.2 (Safety of dynamical systems). Consider a dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ and a set of unsafe states $\mathcal{X}_u \subseteq \mathcal{X}$. Then

- 1) (*Safe controllability*) the system $\mathfrak{S}$ is *safely controllable* with respect to $\mathcal{X}_u$ if, for every initial state $x_0 \in \mathcal{X}_0$, there exists a control sequence $\{u_t\}_{t=0}^\infty \in U$ such that the corresponding trajectory $\{x_t\}_{t=0}^\infty$ starting from x_0 satisfies $\{x_t\}_{t=0}^\infty \cap \mathcal{X}_u = \emptyset$.
- 2) (*Robust safety*) the system $\mathfrak{S}$ is *robustly safe* with respect to $\mathcal{X}_u$ if, for every initial state $x_0 \in \mathcal{X}_0$ and every input sequence $\{u_t\}_{t=0}^\infty \in U$, the corresponding trajectory $\{x_t\}_{t=0}^\infty$ starting from x_0 satisfies $\{x_t\}_{t=0}^\infty \cap \mathcal{X}_u = \emptyset$.

The distinction between safe controllability and robust safety lies in the interpretation of the input U in the dtDS $\mathfrak{S}$. For safe controllability, inputs represent control actions, and the goal is to *design* a policy $\{u_t\}_{t=0}^\infty \in U$ that keeps the closed-loop system safe. For robust safety, inputs represent disturbances, and safety must hold for *all* admissible input sequences $\{u_t\}_{t=0}^\infty \in U$.

2.2 Safety via inductive barrier certificates

A broad class of inductive methods, known as *barrier certificates*, has been developed for safety analysis of dynamical systems [42, 5, 11, 6]. Intuitively, a barrier certificate separates the initial set from the unsafe set and inductively proves that all system trajectories avoid unsafe states. Consider a dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ with unsafe set $\mathcal{X}_u \subseteq \mathcal{X}$. A function $\mathcal{B} : \mathcal{X} \rightarrow \mathbb{R}$ is a *control K -inductive barrier certificate* for $\mathfrak{S}$ with respect to $\mathcal{X}_u$ if there exist $\gamma, \delta, \eta \in \mathbb{R}$ such that $\gamma \leq \delta < \eta$, and

$$\max_{k \in [K-1]} \mathcal{B}(f^k(x, \mathbf{u})) \leq \gamma \quad \forall x \in \mathcal{X}_0, \exists \mathbf{u} \in U^{K-1}, \quad (2)$$

$$\mathcal{B}(x) > \eta \quad \forall x \in \mathcal{X}_u, \quad (3)$$

and, for every $x \in \mathcal{X}$ and every $\mathbf{u} \in U^{K-1}$, there exists $u \in U$ such that

$$\max_{k \in [K-1]} \mathcal{B}(f^k(x, \mathbf{u})) \leq \delta \implies \mathcal{B}(f(f^{K-1}(x, \mathbf{u}), u)) \leq \delta. \quad (4)$$

We refer to the notation section in Appendix B for the definition of $f^k(x, \mathbf{u})$ and $[K-1]$. The corresponding definition of *robust K -inductive barrier certificates* is presented in Appendix C.1. When $K = 1$, control and robust 1-inductive barrier certificates reduce to the classical control and robust barrier certificates studied in the control literature [5, 42]. The following standard result states that the existence of such a certificate guarantees safety. We refer to [42, 5, 39] for a proof.

Theorem 2.3 (Inductive barrier safety certificates). *Consider a dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ with a set of unsafe states $\mathcal{X}_u \subseteq \mathcal{X}$ and an integer $K \in \mathbb{N}_{\geq 0}$. Then $\mathfrak{S}$ is*

- 1) *safely controllable if and only if there exists a control K -inductive barrier function $\mathcal{B} : \mathcal{X} \rightarrow \mathbb{R}$ for $\mathfrak{S}$ with respect to $\mathcal{X}_u$.*
- 2) *robustly safe if and only if there exists a robust K -inductive barrier function $\mathcal{B} : \mathcal{X} \rightarrow \mathbb{R}$ for $\mathfrak{S}$ with respect to $\mathcal{X}_u$.*

Theorem 2.3 reduces robust safety, or safe controllability, to finding a function $\mathcal{B} : \mathcal{X} \rightarrow \mathbb{R}$ satisfying conditions (2)–(4), or conditions (13)–(15). This functional characterization enables the use of computational tools for searching for K -inductive barrier certificates, including SMT solvers [25] and SOS methods [42].

2.3 Monotone dynamical systems

We define the class of monotone systems characterized by order-preserving state-transition maps.

Definition 2.4 (Monotone Dynamical Systems). A dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ is *monotone* if the map $f : \mathcal{X} \times U \rightarrow \mathcal{X}$ satisfies

$$x \leq y \quad \text{and} \quad u \leq v \implies f(x, u) \leq f(y, v).$$

It can be shown that, when the map f is differentiable and the sets $\mathcal{X}$ and U are convex, the discrete-time system $\mathfrak{S}$ is monotone if and only if $Df(x, u) \geq 0_{n \times (n+m)}$ for all $(x, u) \in \mathcal{X} \times U$, where $Df(x, u)$ denotes the Jacobian of f with respect to (x, u) [52].

As discussed in the introduction, monotone systems arise in many natural and engineering applications, either because the underlying physics is *inherently* monotone or because monotonicity is deliberately enforced in *system design* to enable tractable analysis and control. Nevertheless, many important real-world systems are non-monotone due to competitive, inhibitory, or antagonistic interactions among state variables.

Monotone Embeddings. Studying monotone systems entails limited loss of generality. A fundamental observation in monotone dynamical systems is that any dynamical system can be *embedded* into a higher-dimensional monotone system [26, 53, 16, 18] by separating cooperative and competitive interactions among state variables. Thus, although monotonicity is a structural assumption, it is not a narrow one: monotone systems can represent general nonlinear dynamics through suitable embeddings. Consequently, analytical and computational tools for monotone systems can often be extended to broad classes of non-monotone systems via such embeddings.

2.4 Problem statements

In this paper, we study monotone systems with *unknown* state-transition maps. We assume access to a simulator queried at arbitrary state–input pairs to evaluate the transition map. Our goal is to use these queries to establish safe controllability and robust safety, leading to the following two problems.

Problem 1 (Safe control synthesis). *Consider a monotone dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ with an unknown map f and a set of unsafe states $\mathcal{X}_u \subseteq \mathcal{X}$. Given a finite collection of simulator data $\{f(x^i, u^i)\}_{i=1}^N$ corresponding to sampled states $\{x^i\}_{i=1}^N \in \mathcal{X}$ and known sampled inputs $\{u^i\}_{i=1}^N \in U$, design a feedback controller $\kappa : \mathcal{X} \rightarrow U$ such that the states of the closed-loop system*

$$x_{t+1} = f(x_t, \kappa(x_t)), \quad t \geq 0,$$

never enters the unsafe set $\mathcal{X}_u$.

Problem 2 (Robust safety verification). *Consider a monotone dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ with an unknown map f and a set of unsafe states $\mathcal{X}_u \subseteq \mathcal{X}$. Given a finite collection of simulator data $\{f(x^i, u^i)\}_{i=1}^N$ corresponding to sampled states $\{x^i\}_{i=1}^N \in \mathcal{X}$ and unknown inputs $\{u^i\}_{i=1}^N \in U$, determine whether system $\mathfrak{S}$ is robustly safe with respect to $\mathcal{X}_u$.*

Data generation mechanisms. The data acquisition processes in Problems 1 and 2 differ because the inputs play different roles. In safe control synthesis, u^i is a decision variable: for each sampled state x^i , the simulator evaluates $f(x^i, u^i)$ under a *known* chosen control action $u^i \in U$. In robust safety verification, the input represents a disturbance, so for each sampled state x^i , the simulator returns $f(x^i, u^i)$ under an *unknown* disturbance realization $u^i \in U$.

Monotonicity as a structural property. In Problems 1 and 2, monotonicity is assumed without access to the explicit transition map. This assumption is often justified by prior knowledge of the system’s qualitative behavior, inferred, for example, from physical laws or geometric structure. In particular, Definition 2.4 characterizes monotonicity as a *qualitative* property: increasing states or inputs cannot decrease the system response. This property arises naturally in many engineered and natural systems, including traffic networks [17], biological processes [54], and supply chains [60].

3 Monotone inductive barrier certificates

This section identifies a structural property of inductive barrier certificates for monotone systems. We show that, when the unsafe set satisfies a mild order-theoretic condition, inductive barrier certificates can be chosen to inherit the *monotonicity* of the dynamics. This observation simplifies safety verification and control synthesis and enables scalable, data-driven formulations. We now state the result; the proof is given in Appendix D.

Theorem 3.1 (Monotone inductive barrier certificates). *Consider a monotone dtDS $\mathfrak{S} := (\mathcal{X}, \mathcal{X}_0, U, f)$ with an upper closed set¹ of unsafe states $\mathcal{X}_u \subseteq \mathcal{X}$ and an integer $K \in \mathbb{N}_{\geq 1}$. Then*

- 1) (Safe control synthesis) $\mathfrak{S}$ is safely controllable with respect to $\mathcal{X}_u$ if and only if there exists a monotone control K -inductive barrier certificate $\mathcal{B}$ satisfying conditions (2), (3), and (4).
- 2) (Robust safety verification) $\mathfrak{S}$ is robustly safe with respect to $\mathcal{X}_u$ if and only if there exists a monotone robust K -inductive barrier certificate $\mathcal{B}$ satisfying conditions (13), (14), and (15).

Reduction to monotone certificate search. For upper-closed unsafe sets, Theorem 3.1 reduces safe control synthesis, and robust safety verification of monotone systems to the search for *monotone* control and robust, inductive barrier certificates. These are monotone functions satisfying conditions (2)–(4), and conditions (13)–(15). This result is the main theoretical contribution of the paper and forms the foundation of our data-driven approach to safety certification.

Upper closed unsafe sets. The assumption of upper-closed unsafe sets may seem restrictive, since not every unsafe set has this form. However, in many safety-critical systems, unsafe behavior occurs when state variables or aggregate resources exceed thresholds, which naturally yields upper-closed unsafe sets. Examples include drug delivery systems, where excessive chemical concentration can be harmful, and traffic networks, where vehicle density above a critical threshold can cause congestion.

3.1 Data-driven inductive barrier certificates

Although Theorem 3.1 restricts the search for inductive barrier certificates to monotone functions, verifying conditions (2)–(4) (resp. (13)–(15)) still requires either explicit knowledge of the transition map or infinitely many simulator queries over the full state–input space. Building on Theorem 3.1, we exploit certificate monotonicity to construct safety certificates from finite samples while preserving rigorous guarantees. The key idea is to partition the state space and use monotonicity to reduce the infinite-dimensional verification problem to finitely many boundary checks. We partition the state space $\mathcal{X}$ and the input space U into collections of hyper-rectangular cells, denoted by $\{\underline{x}^i, \bar{x}^i\}_{i \in \mathcal{P}}$ and $\{\underline{u}^i, \bar{u}^i\}_{i \in \mathcal{Q}}$, respectively, where $\mathcal{P}$ and $\mathcal{Q}$ are finite index sets associated with these partitions. For given subsets $\mathcal{X}_0 \subseteq \mathcal{X}$ and $\mathcal{X}_u \subseteq \mathcal{X}$, we define index sets $\mathcal{I}$ and $\mathcal{U}$ as the minimal subsets of $\mathcal{P}$

¹A set $\mathcal{S} \subseteq \mathbb{R}^n$ is upper closed if, for every $x \in \mathcal{S}$, any point $y \in \mathbb{R}^n$ satisfying $y \geq x$ also belongs to $\mathcal{S}$.

whose union covers $\mathcal{X}_0$ and $\mathcal{X}_u$, respectively, namely

$$\mathcal{X}_0 \subseteq \bigcup_{i \in \mathcal{I}} [\underline{x}^i, \bar{x}^i], \quad \mathcal{X}_u \subseteq \bigcup_{i \in \mathcal{U}} [\underline{x}^i, \bar{x}^i]. \quad (5)$$

By construction, every state $x \in \mathcal{X}_0$ is component-wise bounded above by $\bar{x}^i$ for some $i \in \mathcal{I}$, while every state $x \in \mathcal{X}_u$ is component-wise bounded below by $\underline{x}^i$ for some $i \in \mathcal{U}$. The search for control barrier certificates that satisfy (2)–(4) can be conservatively reformulated as finding monotone inductive barrier certificates $\mathcal{B} : \mathcal{X} \rightarrow \mathbb{R}$ and monotone feedback controller $\kappa : \mathcal{X} \rightarrow U$ satisfying

$$\max_{k \in [K-1]} \mathcal{B}(f^k(\bar{x}^i, \kappa(\bar{x}^i))) \leq \gamma \quad \forall i \in \mathcal{I}, \quad (6)$$

$$\mathcal{B}(\underline{x}^i) > \eta \quad \forall i \in \mathcal{U}, \quad (7)$$

$$\max_{k \in [K-1]} \mathcal{B}(f^k(\underline{x}^i, \kappa(\underline{x}^i))) \leq \delta \implies \mathcal{B}(f^K(\bar{x}^i, \kappa(\bar{x}^i))) \leq \delta, \quad \forall i \in \mathcal{P} \quad (8)$$

Conditions (6)–(8) reduce certification of inductive barrier certificates over the continuous state set to finitely many maximal and minimal points from the coverings of $\mathcal{X}$ and U . The robust counterpart is given in Appendix C.2. The following result gives a data-driven formal guarantee for robust safety verification and safe control synthesis from finite samples; see Appendix E for the proof.

Theorem 3.2 (Data-driven barrier certificates). *Consider a monotone dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ with unknown map f and a set of unsafe states $\mathcal{X}_u \subseteq \mathcal{X}$. Let $\{[\underline{x}^i, \bar{x}^i]\}_{i \in \mathcal{P}}$ be a family of hyper-rectangles covering the set $\mathcal{X}$, and let $\{[\underline{u}^j, \bar{u}^j]\}_{j \in \mathcal{Q}}$ be a family of hyper-rectangles covering the input space U . Define the index sets $\mathcal{I}$ and $\mathcal{U}$ as the minimal subsets of $\mathcal{P}$ satisfying (5). Then, the system $\mathfrak{S}$*

- 1) (Safe control synthesis) *is safely controllable with respect to $\mathcal{X}_u$ if there exists monotone function $\mathcal{B} : \mathcal{X} \rightarrow \mathbb{R}$ and $\kappa : \mathcal{X} \rightarrow U$ satisfying conditions (6)–(8) with $\gamma \leq \delta < \eta$.*
- 2) (Robust safety verification) *is robustly safe with respect to $\mathcal{X}_u$ if there exists a monotone function $\mathcal{B} : \mathcal{X} \rightarrow \mathbb{R}$ satisfying (16)–(18) with $\gamma \leq \delta < \eta$.*

Data-driven barrier certificates. Theorem 3.2 reduces safety verification over $\mathcal{X}$, namely conditions (2)–(4), to the finite verification problem (6)–(8), evaluated at partition extremal points. Thus, the conditions can be checked using finitely many simulator queries. This reduction exploits the monotone barrier structure of Theorem 3.1 and yields a scalable data-driven certification framework.

Memory consumption. For large K , verifying conditions (6)–(8) can be memory-intensive, since it requires storing intermediate barrier evaluations along trajectories, e.g., $\mathcal{B}(f^1(x, \kappa(x))), \dots, \mathcal{B}(f^K(x, \kappa(x)))$. This overhead becomes significant for long horizons. In Appendix F, we leverage barrier monotonicity to derive a memory-efficient formulation of control, respectively robust, K -inductive barrier certificates.

4 Monotone neural barrier certificates

Theorem 3.2 addresses Problems 1 and 2 using finitely many simulator queries, but does not provide a systematic way to find monotone inductive barrier certificates $\mathcal{B}$ satisfying conditions (6)–(8). In this section, we use *monotone neural networks* to learn such certificates via their approximation power.

4.1 Monotone neural networks

Several monotone neural-network constructions have been proposed and applied across domains [49, 35, 21, 59]. A common architecture uses feedforward networks with nonnegative weights and non-decreasing activations, guaranteeing monotonicity [21]. Despite their simplicity, these networks can have limited expressive power. In particular, with convex activations such as ReLU, the represented functions are themselves convex [38, Lemma 1].

Motivated by these limitations, we introduce *Max-Monotone Networks*, a neural architecture with universal approximation guarantees for monotone functions, tailored to learning inductive barrier certificates satisfying conditions (6)–(8). The construction is inspired by the fact that, under suitable assumptions, such certificates admit a max-separable representation $\mathcal{B}(x) = \max_{i \in \{1, \dots, n\}} h_i(x_i)$, where each $h_i : \mathbb{R} \rightarrow \mathbb{R}$ is monotone. The proof is provided in Appendix H.1.

Despite its simplicity, the max-separable representation is impractical in high dimensions, as it requires a separate monotone one-dimensional network for each coordinate. A natural simplification is to share the component function, yielding $\mathcal{B}(x) = \max_{i \in \{1, \dots, n\}} h(x_i) = h(\max_{i \in \{1, \dots, n\}} x_i)$. This motivates architectures built from coordinate-wise maximization features. However, using only the feature $\max_{i \in \{1, \dots, n\}} x_i$ is too restrictive. To increase expressiveness, we build on classical monotone min-max networks [49] by combining a coordinate-wise maximum feature with multiple monotone feature maps obtained as maxima of monotone affine transformations of the input.

Definition 4.1 (Max-Monotone Networks). A Max-Monotone Network is a mapping $N : \mathbb{R}^n \rightarrow \mathbb{R}$ of the form $N(x) = \min_{j \in \{0, \dots, J\}} N^{(j)}(g^{(j)}(x))$ where, for each $j \in \{0, \dots, J\}$, $N^{(j)} : \mathbb{R} \rightarrow \mathbb{R}$ is a monotone neural network with nonnegative weights and a non-decreasing activation function σ , and the feature map $g^{(j)} : \mathbb{R}^n \rightarrow \mathbb{R}$ is defined by

$$g^{(0)}(x) = \max_{i \in \{1, \dots, n\}} x_i, \quad g^{(j)}(x) = \max_{k \in \{1, \dots, m_j\}} (W_j x + b_j)_k, \quad j \in 1, \dots, J, \quad (9)$$

with matrices $W_j \in \mathbb{R}^{m_j \times n}$ satisfying $W_j \geq 0$.

Appendix H.2 provides a schematic illustration of the Max-Monotone Network architecture. The next theorem establishes the monotonicity and universal approximation properties of this architecture; the proof is referred to Appendix H.3.

Theorem 4.2 (Max-Monotone Networks). *A Max-Monotone Network is monotone. Moreover, for any continuous, bounded, monotone $m : [0, 1]^n \rightarrow \mathbb{R}$ with bounded partial derivatives and any $\varepsilon > 0$, there exists a Max-Monotone Network N such that $|m(x) - N(x)| < \varepsilon$, for all $x \in [0, 1]^n$.*

4.2 Training monotone neural barrier certificates

Ensuring that a neural network satisfies conditions (6)–(8) or (16)–(18) is challenging. We therefore design loss functions for monotone neural networks that explicitly promote these constraints during training. For safe control synthesis, we jointly train two monotone neural networks: a monotone neural barrier certificate $N : \mathcal{X} \rightarrow \mathbb{R}$ parameterized as in Definition 4.1, and a monotone feedback policy $\kappa : \mathcal{X} \rightarrow U$ with the same architecture except that each branch outputs a vector in $\mathbb{R}^m$ and the final minimum is applied componentwise across branches. Both networks are trained with the squared ReLU-error loss $L_{\text{total}}^s := L_1^s + L_2^s + L_3^s$. The terms L_1^s , L_2^s , and L_3^s enforce the data-driven barrier constraints (6), (7), and (8), respectively:

$$L_1^s = \sum_{i \in \mathcal{I}} [\text{ReLU}(\max_{k \in [K-1]} (N(f^k(\bar{x}^i, \kappa(\bar{x}^i))) - \gamma))^2], \quad (10)$$

$$L_2^s = \sum_{i \in \mathcal{U}} [\text{ReLU}(\eta - N(\underline{x}^i))]^2, \quad (11)$$

$$L_3^s = \sum_{i \in \mathcal{P}} r_i [\text{ReLU}(N(f^K(\bar{x}^i, \kappa(\bar{x}^i))) - \delta)]^2, \quad (12)$$

where $r_i = \text{sign}(\text{ReLU}(-\max_{k \in [K-1]} (N(f^k(\bar{x}^i, \kappa(\bar{x}^i))) - \delta))$, for every $i \in \mathcal{P}$. The parameters $\gamma, \delta, \eta \in \mathbb{R}$, with $\gamma \leq \delta < \eta$, are tunable hyperparameters that act as margins to improve the robustness of the barrier certificate. Training uses gradient-based optimization of L_{total}^s and terminates once $L_{\text{total}}^s = 0$. The robust safety verification loss formulation is given in Appendix C.3.

Safety certificates from zero loss. Achieving zero loss, $L_{\text{total}} = 0$, is equivalent to satisfying the inductive barrier conditions (6)–(8). Thus, the learned monotone neural network N is an inductive barrier certificate that formally certifies system safety; see Theorem I.1 in Appendix I.

Neural network loss functions. The total loss $L_{\text{total}} = L_1 + L_2 + L_3$ can be adjusted to improve training, by replacing nonsmooth terms such as ReLU or sign with smoother MSE objectives, while using the original barrier loss only for termination. In our experiments, however, we directly optimize the proposed loss and use it for termination, observing stable training and strong performance.

Iterative refinement of the hyper-rectangular cover. While zero training loss yields a formal safety certificate, convergence to zero is not guaranteed and may be limited by insufficient samples. This motivates adaptive sampling algorithms that iteratively refine the partition of the state space, enriching the dataset with informative samples and improving the chance of certification. Appendix J.4 presents

two refinement strategies: *monotone shrinking refinement* (Algorithm 3) and *arbitrary refinement* (Algorithms 4–5). Algorithm 1 (resp. Algorithm 2) integrates these refinements into training Max-Monotone Networks with the losses (10)–(12) (resp. losses (19)–(21)). The following theorem establishes soundness and sample complexity bounds for Algorithm 1; see Appendix J.4 for the proof.

Theorem 4.3 (Soundness and linear simulator-query complexity of Algorithm 1). *Consider a monotone dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ with a set of unsafe states $\mathcal{X}_u \subseteq \mathcal{X} \subseteq \mathbb{R}^n$. Let $\{[\underline{x}^i, \bar{x}^i]\}_{i \in \mathcal{P}}$ be a family of hyper-rectangles covering $\mathcal{X}$, and let $\{[\underline{u}^j, \bar{u}^j]\}_{j \in \mathcal{Q}}$ be a family of hyper-rectangles covering U . Define the index sets $\mathcal{I}$ and $\mathcal{U}$ as the minimal subsets of $\mathcal{P}$ satisfying (5) and suppose that the cardinality of $\mathcal{P}, \mathcal{I}, \mathcal{U}$ are independent of n . Then:*

- 1) (Soundness) *If Algorithm 1 returns $(N, \kappa, L_{\text{total}})$, then the system $\mathfrak{S}$ is safely controllable with respect to $\mathcal{X}_u$ using the controller κ .*
- 2) (Sample complexity) *If Algorithm 3 is used as the refinement routine in line 10 of Algorithm 1, then the total number of simulator queries required by Algorithm 1 is $\mathcal{O}(n)$.*

For high-dimensional systems, Algorithm 1 may require many samples to construct a valid monotone barrier certificate. The inductive parameter K helps mitigate this issue. Rather than relying only on finer state-space partitions, larger K values incorporate multi-step system evolution into the certification conditions, reducing the partition resolution needed for certification and lowering sample complexity. Appendix K demonstrates the practical benefits of tuning K under a fixed partition.

Algorithm 1 Monotone neural control barrier learning with adaptive refinement

Inputs: Simulator access to the dynamics f ; hyper-rectangular covers $\{[\underline{x}^i, \bar{x}^i]\}_{i \in \mathcal{P}}$, $\{[\underline{x}_0^i, \bar{x}_0^i]\}_{i \in \mathcal{I}}$, and $\{[\underline{x}_u^j, \bar{x}_u^j]\}_{j \in \mathcal{U}}$, of $\mathcal{X}$, $\mathcal{X}_0$, and $\mathcal{X}_u$, respectively; margins $\gamma < \delta < \eta$; maximum inductive parameter K_{max} ; maximum shrinking budget $n_{\text{max}}^{\mathcal{P}}$; and maximum number of epochs N_{ep} .

Output: A monotone neural control barrier certificate N , controller κ , and final loss L_{total} , if found; otherwise FAIL and final loss L_{total} .

```

1: Initialize  $\mathcal{H} \leftarrow \{[\underline{x}^\ell, \bar{x}^\ell]\}_{\ell \in \mathcal{P} \setminus \mathcal{U}}$ ,  $\mathcal{C} \leftarrow \emptyset$ ,  $L_{\text{total}} \leftarrow \infty$ .
2: for  $p = 1, \dots, n_{\text{max}}^{\mathcal{P}}$  do
3:   for  $K = 1, \dots, K_{\text{max}}$  do
4:     Define  $L_{\text{shrink}} = \sum_{x^c \in \mathcal{C}} [\text{ReLU}(\eta - N(x^c))]^2$ .
5:     Train  $N$  and  $\kappa$  for at most  $N_{\text{ep}}$  epochs using  $L_{\text{total}} = L_1 + L_2 + L_3 + L_{\text{shrink}}$ , where
        $L_1, L_2, L_3$  are defined in (10)–(12) using  $\mathcal{I}, \mathcal{U}$ , the current unresolved index set induced
       by  $\mathcal{H}$ , the current value of  $K$ , and margins  $\gamma, \delta, \eta$ .
6:     if  $L_{\text{total}} = 0$  then
7:       return  $N, \kappa, L_{\text{total}}$ 
8:     end if
9:   end for
10:  Apply either Algorithm 3 or Algorithm 4 to obtain refined collections  $(\mathcal{H}^+, \mathcal{C}^+)$ 
11:  if the chosen refinement algorithm returns FAIL then
12:    return FAIL,  $L_{\text{total}}$ 
13:  else
14:    Update  $(\mathcal{H}, \mathcal{C})$  using the returned  $(\mathcal{H}^+, \mathcal{C}^+)$ 
15:  end if
16: end for
17: return FAIL,  $L_{\text{total}}$ 

```

5 Numerical experiments

In this section, we demonstrate the computational efficiency and scalability of the proposed framework across several representative benchmark systems. The dynamics are provided for clarity but are not used during learning; each system is treated as a black box, using only sampled data. We compare with two baselines: the counterexample-guided approach of [25] and the Lipschitz-based method of [7]. Additional implementation details, hardware specifications, neural architectures, and baseline configurations are given in Appendix N. Table 1 shows scalability on Lotka–Volterra systems from 2 to 100 dimensions, while Table 2 compares runtime and sample complexity across case studies. The results show that competing methods either fail to converge or require exponentially many samples as dimension grows, whereas our approach remains scalable and sample-efficient in high dimensions.

Table 1: Safety-verification performance on Lotka–Volterra with increasing dimension. Averaged over 10 runs. NC: no convergence within 12 hours; EM: out of memory. Our method uses $K = 2$.

Dim	[25]		[7]		Ours	
	Time	Samples	Time	Samples	Time	Samples
2	47.5s	1000	85.9s	10^6	5.2s	9
3	NC	1000	NC	$2^6 \times 10^6$	6.2s	10
5	NC	1000	NC	$2^6 \times 10^6$	5.8s	12
10	NC	1000	NC	$2^6 \times 10^6$	5.9s	17
50	NC	1000	EM	$\Omega(2^{50})$	6.6s	57
100	NC	1000	EM	$\Omega(2^{100})$	10.1s	107

Table 2: Runtime and sample complexity across case studies (averaged over 10 runs). NC indicates failure to converge within 12 hours, and EM indicates exceeding available memory.

Case (Dim)	Time (s)			Samples		
	[25]	[7]	Ours	[25]	[7]	Ours
Freeway traffic (1,000D)	NC	EM	542.2	10^3	$\Omega(2^{1000})$	151,001
Kuarmoto (all-to-all) (9,000D)	NC	EM	395.6	10^3	$\Omega(2^{9000})$	9,007
IEEE 118-Bus (118D)	NC	EM	9.7	10^3	$\Omega(2^{118})$	125
IEEE 2383WP (2,383D)	NC	EM	30.4	10^3	$\Omega(2^{2383})$	2,390
PEGASE (13,659D)	NC	EM	787.1	10^3	$\Omega(2^{13659})$	13,666
Lotka–Volterra (2,000D)	NC	EM	62.3	10^3	$\Omega(2^{2000})$	2,301
Ribosome Flow Model (500D)	NC	EM	3712.1	10^3	$\Omega(2^{500})$	8,001
Ribosome Flow Model (1,000D)	NC	EM	14112.5	10^3	$\Omega(2^{1000})$	16,001

Freeway traffic flow dynamics. We consider the Daganzo cell transmission model (CTM) [19], which captures demand and supply constraints on a line-digraph traffic network (Fig. 5). This model induces a monotone system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ [13, Lemma 4.28], with $\mathcal{X} = [0, 50]^n$, $\mathcal{X}_0 = [0, 5]^n$, and $U = [0, 7.5]$. The unsafe set is $\mathcal{X}_u = [0, 50]^n \setminus [0, 40]^n$. Our framework certifies safety for $n = 1000$, $K = 50000$, using 151,001 samples and trains a monotone neural robust barrier certificate in 542.2 s on average over 10 runs. Details are in Appendix N.2.

Network of coupled oscillators. We consider discrete-time Kuramoto oscillator networks [51] with all-to-all topology and with topologies from IEEE power-system testbeds from MATPOWER black-box model [66], yielding monotone systems $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$. The state space is $\mathcal{X} = [0, \frac{\pi}{4}]^n$, the initial set is $\mathcal{X}_0 = [0, \frac{\pi}{16}]^n$, and the input set is $U = [-\frac{\pi}{4}, \frac{\pi}{4}]^n$. The unsafe set is $\mathcal{X}_u = [0, \frac{\pi}{4}]^n \setminus [0, \frac{\pi}{8}]^n$. For all-to-all graph structure, we certify safety for $n = 9000$, $K = 2$, using 9,007 samples and synthesizing a safe control certificate in 395.6 s on average over 10 runs. For IEEE-based topologies, we consider IEEE-118 ($n = 118$), IEEE-2383WP ($n = 2383$), and IEEE-13659-PEGASE ($n = 13659$), synthesizing safe control barrier certificates in 9.7, 30.4, and 787.1 s, respectively, with $K = 2$. Further details are in Appendix N.3.

Population dynamics. We consider a discrete-time Lotka–Volterra population model [13], which induces a monotone system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$ on the nonnegative orthant. The state space is $\mathcal{X} = [0, 1]^n$, the initial set is $\mathcal{X}_0 = [0, 0.4]^n$, and the unsafe set is $\mathcal{X}_u = [0, 1]^n \setminus [0, 0.5]^n$. Our framework certifies safety for $n = 2000$, $K = 100$, using 2,301 samples and learning a monotone neural robust barrier certificate in 62.3 s on average over 10 runs. Further details are in Appendix N.4.

Ribosome Flow Model. We consider a Ribosome Flow Model (RFM) [3, 36] with n sites, inducing a monotone system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$. The state space is $\mathcal{X} = [0, 1]^n$, the initial set is $\mathcal{X}_0 = [0, 0.2]^n$, and the unsafe set is $\mathcal{X}_u = [0, 1]^n \setminus [0, 0.9]^n$. Our framework certifies safety for $n = 500$, $K = 2500$, using 8,001 samples and learning a valid barrier certificate in 3712.1 s over 10 runs. For $n = 1000$, it certifies safety with $K = 5000$, using 16,001 samples in 14122.5 s. Details are in Appendix N.5.

6 Conclusions.

This paper introduced a scalable, learning-based framework for synthesizing safety controllers for high-dimensional unknown *monotone* systems directly from data. We showed that barrier certificates for monotone systems inherit the underlying monotonicity structure of the dynamics, thereby reducing safety verification to a finite collection of localized boundary checks. Building on this structural property, we proposed a specialized neural network architecture, called *Max-Monotone Networks*, for learning such barrier certificates directly from data. Our approach yields substantially improved sample efficiency over existing black-box methods and can avoid exponential growth in sample complexity. Experiments on large-scale benchmarks further show that our method scales well in both data usage and training stability, even in high-dimensional settings.

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A Related Works

Barrier Certificates. The work [42] introduced barrier certificates (BCs) as functions whose level sets over-approximate system reachable sets. Control barrier certificates (CBCs) later emerged as a key tool for synthesizing safe controllers [57, 29, 15, 20], typically using Sum-of-Squares (SOS) optimization [43, 65, 47].

Neural Barrier Certificates. The paper [63] introduced neural network-based barrier certificates, using a shallow network and MILP for formal verification. Later, [40] employed SMT solvers to find counterexamples for iterative refinement. Neural barrier certificates have since been extended to hybrid systems [64] and stochastic dynamics [37].

Neural Control Barrier Certificates. Neural control barrier certificates (NCBCs) extend these ideas to safe controller synthesis. Recent work [45, 32, 23, 22, 34, 33] learns both barrier functions and control policies using neural networks. Verification typically relies on SMT solvers [1, 63, 25], reachable set methods [56], polynomial approximations [48], Lipschitz bounds [7], or ReLU verification [30]. Recently, [62, 61] an exact NCBC verification algorithm was proposed that synthesizes a controller if the barrier is valid. Most of these methods, however, assume full system knowledge (except [7]) and rely heavily on SMT solvers, which scale poorly to deep networks due to their exponential complexity.

B Mathematical Notations

We denote the sets of real, nonnegative real, and positive real numbers by $\mathbb{R}$, $\mathbb{R}_{\geq 0}$, and $\mathbb{R}_{> 0}$, respectively, and the sets of nonnegative integers and positive integers by $\mathbb{N}$ and $\mathbb{N}_{> 0}$, respectively. Given an integer $K \in \mathbb{N}_{> 0}$, we define $[K] = \{0, 1, \dots, K\}$. Let $\mathcal{X} \subseteq \mathbb{R}^n$ and $f : \mathcal{X} \rightarrow \mathcal{X}$. For $i \in \mathbb{N}$, we define $f^i : \mathcal{X} \rightarrow \mathcal{X}$ recursively by $f^i(x) = f(f^{i-1}(x))$, for every $i \in \mathbb{N}_{> 0}$ and $f^0(x) = x$, for every $x \in \mathcal{X}$. Let $\mathcal{X} \subseteq \mathbb{R}^n$, $U \subseteq \mathbb{R}^m$, and $f : \mathcal{X} \times U \rightarrow \mathcal{X}$. For an input sequence $\mathbf{u} = (u_0, \dots, u_{K-1}) \in U^K$, we define the iterates of f recursively as

$$\begin{aligned} f^0(x, \mathbf{u}) &= x, & \forall x \in \mathcal{X} \\ f^{i+1}(x, \mathbf{u}) &= f(f^i(x, \mathbf{u}), u_i), & \forall x \in \mathcal{X}, \forall i \in [K-1]. \end{aligned}$$

For sets A and $B \subseteq A$, we use $A \setminus B$ for set difference, $A \times B$ for Cartesian product, and write $|A|$ for the cardinality of A . For $k \in \mathbb{N}_{\geq 0}$, we denote by A^k the k -fold Cartesian product of the set A with itself. For a vector $x = (x_1, \dots, x_n)^\top \in \mathbb{R}^n$, we denote its i th component by x_i . We write $x \leq y$ to indicate element-wise inequality, i.e., $x_i \leq y_i$ for all $i \in \{1, \dots, n\}$. The Euclidean norm and the ℓ_1 -norm of x are denoted by $\|x\| = \sqrt{\sum_{i=1}^n x_i^2}$ and $\|x\|_1 = \sum_{i=1}^n |x_i|$, respectively. A hyper-rectangle is represented by $[\underline{x}, \bar{x}] = \{x \in \mathbb{R}^n \mid \underline{x} \leq x \leq \bar{x}\}$. A set $S \subseteq \mathbb{R}^n$ is *lower closed* if $x \in S$ and $y \leq x$ imply $y \in S$, and *upper closed* if $x \in S$ and $y \geq x$ imply $y \in S$. Given a set $A \subseteq \mathbb{R}^n$, a point $x \in A$ is *maximal* (resp. *minimal*) if there is no $y \in A$ such that $x \leq y$ (resp. $y \leq x$). Let $\mathcal{X} \subseteq \mathbb{R}^n$. A map $f : \mathcal{X} \rightarrow \mathcal{Y}$ is *monotone* if, for all $x, y \in \mathcal{X}$ such that $x \leq y$, we have $f(x) \leq f(y)$. The rectified linear unit (ReLU) is defined by $\text{ReLU}(x) = \max(0, x)$. Throughout the paper, $0_{n \times n}$ denotes the $n \times n$ zero matrix, and $\mathbf{1}_n$ denotes the n -dimensional column vector of ones.

C Robust inductive barrier certificates

In this appendix, we present formulas for robust K -inductive barrier certificates.

C.1 Definition of Robust K -inductive barrier certificates

A function $\mathcal{B} : \mathcal{X} \rightarrow \mathbb{R}$ is a *robust K -inductive barrier certificate* for $\mathfrak{S}$ with respect to $\mathcal{X}_u$ if there exist $\gamma, \delta, \eta \in \mathbb{R}_{\geq 0}$ such that $\gamma \leq \delta < \eta$ and, for every $\mathbf{u} \in U^K$, the following conditions hold:

$$\max_{k \in [K-1]} \mathcal{B}(f^k(x, \mathbf{u})) \leq \gamma \quad \forall x \in \mathcal{X}_0, \quad (13)$$

$$\mathcal{B}(x) \geq \eta \quad \forall x \in \mathcal{X}_u, \quad (14)$$

$$\max_{k \in [K-1]} \mathcal{B}(f^k(x, \mathbf{u})) \leq \delta \implies \mathcal{B}(f^K(x, \mathbf{u})) \leq \delta, \quad \forall x \in \mathcal{X}. \quad (15)$$

C.2 Data-driven robust barrier certificate formulation

By construction, every state $x \in \mathcal{X}_0$ is component-wise bounded above by $\bar{x}^i$ for some $i \in \mathcal{I}$, while every state $x \in \mathcal{X}_u$ is component-wise bounded below by $\underline{x}^i$ for some $i \in \mathcal{U}$. Thus, the search for robust barrier certificates satisfying (13)–(15) can be conservatively reformulated as finding a monotone function $\mathcal{B} : \mathcal{X} \rightarrow \mathbb{R}$ such that, for every $\bar{\mathbf{u}} \in \left(\bigcup_{j \in \mathcal{Q}} \{\bar{u}^j\}\right)^K$ and $\underline{\mathbf{u}} \in \left(\bigcup_{j \in \mathcal{Q}} \{\underline{u}^j\}\right)^K$,

$$\max_{k \in [K-1]} \mathcal{B}(f^k(\bar{x}^i, \bar{\mathbf{u}})) \leq \gamma \quad \forall i \in \mathcal{I}, \quad (16)$$

$$\mathcal{B}(\underline{x}^i) \geq \eta \quad \forall i \in \mathcal{U}, \quad (17)$$

$$\max_{k \in [K-1]} \mathcal{B}(f^k(\underline{x}^i, \underline{\mathbf{u}})) \leq \delta \Rightarrow \mathcal{B}(f^K(\bar{x}^i, \bar{\mathbf{u}})) \leq \delta, \quad \forall i \in \mathcal{P} \quad (18)$$

Conditions (16)–(18) reduce the verification of barrier certificates from the continuous state set to a finite set of carefully selected maximal and minimal points derived from the coverings of $\mathcal{X}$ and U .

C.3 Robust safety verification loss

For the robust safety verification problem (Problem 2), we train a monotone neural network $N : \mathcal{X} \rightarrow \mathbb{R}$ as a monotone robust barrier certificate using a squared ReLU-error loss. The total loss is defined as $L_{\text{total}}^v = L_1^v + L_2^v + L_3^v$, where each term corresponds to one of the barrier certificate conditions.

The individual loss terms are given by

$$L_1^v = \sum_{i \in \mathcal{I}, j \in \mathcal{Q}} [\text{ReLU}(\max_{k \in [K-1]} (N(f^k(\bar{x}^i, \bar{u}^j)) - \gamma)]^2, \quad (19)$$

$$L_2^v = \sum_{i \in \mathcal{U}} [\text{ReLU}(\eta - N(\underline{x}^i))]^2, \quad (20)$$

$$L_3^v = \sum_{i \in \mathcal{P}, j \in \mathcal{Q}} r_{ij} [\text{ReLU}(N(f^K(\bar{x}^i, \bar{u}^j)) - \delta)]^2, \quad (21)$$

where

$$r_{ij} = \text{sign} \left(\text{ReLU} \left(- \max_{k \in [K]} (N(f^k(\underline{x}^i, \underline{u}^j)) - \delta) \right) \right).$$

The parameters $\gamma, \delta, \eta \in \mathbb{R}$ satisfy $\gamma \leq \delta < \eta$ and serve as robustness margins.

D Proof of Theorem 3.1

We start by proving Theorem 3.1 part 1. We divide the proof into **if** part and **only if** part.

Proof of if part. Suppose that $B : \mathcal{X} \rightarrow \mathbb{R}$ is a monotone barrier certificate satisfying (2), (3), and (4). Then, by condition (2), for any $x_0 \in \mathcal{X}_0$, there exists $\mathbf{u} = (u_0, \dots, u_{K-2}) \in U^{K-1}$ such that $B(f^k(x_0, \mathbf{u})) \leq \gamma$, for every $k \in [K-1]$. Since $\gamma \leq \delta$, this means that $B(f^k(x_0, \mathbf{u})) \leq \delta$, for every $k \in [K-1]$. Using condition (4), we deduce that there exists $u_{K-1} \in U$ such that

$$B(f(f^{K-1}(x_0, \mathbf{u}), u_{K-1})) \leq \delta.$$

Now consider the point $x_1 = f(x_0, u_0) \in \mathcal{X}$ and the control sequence $\hat{\mathbf{u}} = (u_1, \dots, u_{K-1}) \in U^K$. It is clear that, $\mathbb{B}(f^k(x_1, \hat{\mathbf{u}})) = \mathbb{B}(f^{k+1}(x_0, \mathbf{u}))$, for every $k \in [K-1]$. Therefore, for every $k \in [K-1]$,

$$B(f^k(x_1, \hat{\mathbf{u}})) \leq \delta.$$

Using condition (4), we deduce that there exists $u_K \in U$ such that

$$B(f(f^K(x_0, \mathbf{u}), u_K)) = \mathbb{B}(f(f^{K-1}(x_1, \hat{\mathbf{u}}), u_K)) \leq \delta.$$

Thus, by iteratively applying this procedure, we can find a sequence $\{u_t\}_{t=0}^\infty$ such that, for every $t \geq 0$, we have

$$B(x_{t+1}) \leq \delta.$$

where $x_{t+1} = f(x_t, u_t)$, for every $t \geq 0$. This implies that $\{x_t\}_{t=0}^\infty \subseteq \{y \in \mathbb{R}^n \mid \mathbb{B}(y) \leq \delta\}$. On the other hand, by condition (3), we know that, for every $x \in \mathcal{X}_u$, we have $B(x) \geq \eta > \delta$. As a result $\{x_t\}_{t=0}^\infty \cap \mathcal{X}_u = \emptyset$. This means that the system $\mathcal{S}$ is safe with the feedback controller $\kappa : \mathcal{X} \rightarrow \mathbb{R}$.

Proof of only if part. Suppose that the system $\mathfrak{S}$ is safely controllable with respect to $\mathcal{X}_u$. Before we state the proof, we introduce several notations and intermediate results that will be used throughout the proof. Since $\mathfrak{S}$ is safely controllable with respect to $\mathcal{X}_u$, for every $x_0 \in \mathcal{X}$, there exists a control sequence $\{u_t\}_{t=0}^\infty$ such that the associated trajectory $\{x_t\}_{t=0}^\infty$ avoids the unsafe set $\mathcal{X}_u$. We label this specific control sequence associated to $x_0 \in \mathcal{X}_0$ by $\{u_t^{x_0}\}_{t=0}^\infty$. The *reachable set* of $\mathfrak{S}$, which is denoted by $\Omega \subseteq \mathcal{X}$, is defined by

$$\Omega = \bigcup_{T \in \mathbb{Z}_{\geq 0}} \Omega_T, \quad (22)$$

$$\Omega_T = \{x_T \in \mathcal{X} \mid \{x_t\}_{t=0}^\infty \text{ is a traj of } \mathfrak{S} \text{ starting from } x_0 \in \mathcal{X}_0 \text{ with controller } \{u_t^{x_0}\}_{t=0}^\infty\}.$$

Motivated by the definition of the reachable set of $\mathfrak{S}$ in (22) we define the set $\widehat{\Omega} \subseteq \mathcal{X}$ as the smallest lower closed set in $\mathcal{X}$ containing Ω . More specifically, $\widehat{\Omega}$ is defined as:

$$\widehat{\Omega} = \{x \in \mathcal{X} \mid x \leq y, \text{ for some } y \in \Omega\}. \quad (23)$$

Given $\gamma, \eta \in \mathbb{R}_{\geq 0}$ such that $\gamma \leq \eta$, we use the lower closed set $\widehat{\Omega}$ to define the function $B : \mathcal{X} \rightarrow \mathbb{R}$ as follows:

$$B(x) = \begin{cases} \gamma & x \in \widehat{\Omega} \\ \eta + 1 & x \notin \widehat{\Omega} \end{cases} \quad (24)$$

The next lemma proves a useful property of the function B .

Lemma D.1. *Consider a monotone dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ with the lower closed set $\widehat{\Omega}$ as defined in (27). Then the function $B : \mathcal{X} \rightarrow \mathbb{R}$ defined in (24) is monotone.*

Proof. Let $x, y \in \mathcal{X}$ be such that $x \leq y$. We consider two cases: i) $y \in \widehat{\Omega}$, and ii) $y \notin \widehat{\Omega}$.

1. In this case, there exists $z \in \Omega$ such that $y \leq z$. Note that we have $x \leq y \leq z$. This implies that $x \in \widehat{\Omega}$. As a result, by definition of B in (24), we have $B(x) = B(y) = \gamma$.
2. In this case, by definition of B in (24), we have $B(y) = \eta + 1$. This means that $B(x) \leq \eta + 1 = B(y)$.

In both cases i) and ii), we have $B(x) \leq B(y)$. This implies that B is a monotone map. $\square$

Now we get back to Theorem 3.1 part 1.

Let the lower closed set $\widehat{\Omega}$ be defined in (23) and the function $B : \mathcal{X} \rightarrow \mathbb{R}$ be defined as in (24). Our goal is to show that $B : \mathcal{X} \rightarrow \mathbb{R}$ is a monotone K -inductive control barrier certificate for $\mathfrak{S}$ satisfying conditions (2), (3), and (4). By Lemma D.1, it is clear that B is monotone.

1. (**B satisfies condition (2)**) Suppose that $x_0 \in \mathcal{X}_0$. Since $\mathfrak{S}$ is safely controllable with respect to $\mathcal{X}_u$, there exists a controller $\{u_t^{x_0}\}_{t=0}^\infty$ such that system $\mathfrak{S}$ under this controller starting from x_0 never enters $\mathcal{X}_u$. Let $\mathbf{u} = (u_0^{x_0}, u_1^{x_0}, \dots, u_{K-1}^{x_0})$. By the definition of Ω , this implies that $x_0 \in \Omega$. Moreover, $f^k(x_0, \mathbf{u}) \in \Omega$, for every $k \in [K-1]$. As a result, we have $f^k(x_0, \mathbf{u}) \in \Omega \subseteq \widehat{\Omega}$, for every $k \in [K-1]$. Using the definition of B in (24), we get $B(f^k(x_0, \mathbf{u})) \leq \gamma$, for every $k \in [K-1]$. This shows that B satisfies condition (2) with the control input $\mathbf{u}$.
2. (**B satisfies condition (3)**) Suppose that $x \in \mathcal{X}_u$. Since $\mathfrak{S}$ is safe with respect to $\mathcal{X}_u$, we get that $x \notin \Omega$. Note that $\widehat{\Omega}$ is the smallest lower closed set containing Ω and $\mathcal{X}_u$ is an upper closed set. As a result, $x \notin \widehat{\Omega}$. By definition of B in (24), we have $B(x) = \eta + 1 > \gamma$. This shows that B satisfies condition (3).
3. (**B satisfies condition (4)**) Let $x \in \mathcal{X}$ and the control input $\mathbf{u} = (u_0, \dots, u_{K-2}) \in U^{K-1}$ is such that $B(f^k(x, \mathbf{u})) \leq \delta$, for every $k \in [K-1]$. By definition of B and using the fact that $\delta < \eta$, we have $f^k(x, \mathbf{u}) \in \widehat{\Omega}$, for every $k \in [K-1]$. This means that, there exists $y \in \Omega$ such that $f^{K-1}(x, \mathbf{u}) \leq y$. This implies that, there exists $T > 0$, a point $x_0 \in \mathcal{X}_0$

and an associated controller $\{u_t^{x_0}\}_{t=0}^\infty$ such that $y = x_T$, where $x_{t+1} = f(x_t, u_t^{x_0})$, for every $t \geq 0$. Since f is monotone, we get

$$f(f^{K-1}(x, \mathbf{u}), u_T^{x_0}) \leq f(y, u_T^{x_0}) = f(x_T, u_T^{x_0}) = x_{T+1} \in \Omega_{T+1} \subseteq \Omega \subseteq \widehat{\Omega}.$$

Therefore $B(f(f^{K-1}(x, \mathbf{u}), u_T^{x_0})) = \gamma \leq \delta$. This shows that B satisfies condition (4).

This completes the proof of Theorem 3.1 part 1.

Now we focus on Theorem 3.1 part 2. Before proving Theorem 3.1 part 2, we introduce several notations and intermediate results that will be used throughout the proof. Consider a monotone dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$. The *reachable set* of $\mathfrak{S}$, denoted by $\Omega \subseteq \mathcal{X}$, is defined as follows:

$$\Omega = \bigcup_{T \in \mathbb{Z}_{\geq 0}} \Omega_T, \quad (25)$$

$$\Omega_T = \{x_T \in \mathcal{X} \mid \{x_t\}_{t=0}^\infty \text{ is a traj of } \mathfrak{S} \text{ with some input } \{u_t\}_{t=0}^\infty \text{ with } x_0 \in \mathcal{X}_0\}.$$

We first state the following intermediate result about the reachable set Ω .

Lemma D.2. *Consider a monotone dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ with the reachable set Ω defined in (25). Then, for every $x \in \Omega$ and every $u \in U$, we have $f(x, u) \in \Omega$.*

Proof. Let $x \in \Omega$ and $u \in U$. Then there exists $T \in \mathbb{Z}_{\geq 0}$ such that $x \in \Omega_T$. This means that there exists $\{x_t\}_{t=0}^\infty$ a trajectory of $\mathfrak{S}$ with the input $\{u_t\}_{t=0}^\infty$ starting from $x_0 \in \mathcal{X}_0$ such that $x = x_T$. Now consider the input $\{w_t\}_{t=0}^\infty$ defined by

$$w_t = \begin{cases} u_t & t \leq T, \\ u & t > T. \end{cases} \quad (26)$$

Let $\{y_t\}_{t=0}^\infty$ be the trajectory of $\mathfrak{S}$ with the input $\{w_t\}_{t=0}^\infty$ starting from x_0 . By definition of the input $\{w_t\}_{t=0}^\infty$ in (26), it is clear that $y_t = x_t$, for every $t \leq T$. Moreover, for $t = T + 1$, we have

$$y_{T+1} = f(y_T, w_T) = f(y_T, u) = f(x_T, u) = f(x, u),$$

where the second equality holds by definition of the input $\{w_t\}_{t=0}^\infty$ in (26), the third equality holds using the fact that $x_t = y_t$, for every $t \leq T$, and the fourth equality holds because $x = x_T$. Therefore, by definition of Ω_{T+1} , we have $f(x, u) \in \Omega_{T+1}$ and thus $f(x, u) \in \Omega$. $\square$

Motivated by the definition of the reachable set of $\mathfrak{S}$ in (25) we define the set $\widehat{\Omega} \subseteq \mathcal{X}$ as the smallest lower closed set in $\mathcal{X}$ containing Ω . More specifically, $\widehat{\Omega}$ is defined as:

$$\widehat{\Omega} = \{x \in \mathcal{X} \mid x \leq y, \text{ for some } y \in \Omega\}. \quad (27)$$

Given $\gamma, \eta \in \mathbb{R}_{\geq 0}$ such that $\gamma \leq \eta$, we use the lower closed set $\widehat{\Omega}$ to define the function $B : \mathcal{X} \rightarrow \mathbb{R}$ as follows:

$$B(x) = \begin{cases} \gamma & x \in \widehat{\Omega} \\ \eta + 1 & x \notin \widehat{\Omega} \end{cases} \quad (28)$$

The next lemma proves a useful property of the function B .

Lemma D.3. *Consider a monotone dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ with the lower closed set $\widehat{\Omega}$ as defined in (27). Then the function $B : \mathcal{X} \rightarrow \mathbb{R}$ defined in (28) is monotone.*

Proof. Let $x, y \in \mathcal{X}$ be such that $x \leq y$. We consider two cases i) $y \in \widehat{\Omega}$ and ii) $y \notin \widehat{\Omega}$.

- i) In this case, by definition of $\widehat{\Omega}$ in (27), there exists a point $z \in \Omega$ such that $y \leq z$. Moreover, we have $x \leq y \leq z$. Therefore, we have $x \in \widehat{\Omega}$ and $B(x) = \gamma = B(y)$.
- ii) In this case, we have $B(y) = \eta + 1$. Since we know that $\gamma \leq \eta < \eta + 1$, we get $B(x) \leq \eta + 1 = B(y)$.

Thus, in both cases i) and ii), we have $B(x) \leq B(y)$. This implies that B is a monotone map. $\square$

Now we go back to the proof of Theorem 3.1 part 2. We divide the proof into **if** part and **only if** part.

Proof of if part. Suppose that $B : \mathcal{X} \rightarrow \mathbb{R}$ is a monotone function satisfying conditions (13), (14), and (15). This means that $B : \mathcal{X} \rightarrow \mathbb{R}$ defined in (28) is a monotone K -inductive robust barrier certificate for $\mathfrak{S}$. Let $x_0 \in \mathcal{X}_0$. Then, by condition (13), $f^k(x_0, \mathbf{u}) \in \Omega \subseteq \widehat{\Omega}$, for every $k \in [K-1]$ and every $\mathbf{u} \in U^K$. Using definition of B in (28), we get that $B(f^k(x_0, \mathbf{u})) \leq \gamma$, for every $k \in [K-1]$ and every $\mathbf{u} \in U^K$. Now, we can use condition (14) of the barrier to deduce that $B(f^K(x_0, \mathbf{u})) \leq \gamma$, for every $\mathbf{u} \in U^{K+1}$. By recursively applying condition (14), we obtain that $B(f^t(x_0, \mathbf{u})) \leq \gamma$, for every $t \in \mathbb{N}$ and every $\mathbf{u} \in U^{k+1}$. Note that, for every $x \in \mathcal{X}_u$, we have $B(x) > \gamma \geq \eta$. This implies that, for every $t \in \mathbb{N}$ and every $\mathbf{u} \in U^{k+1}$, we have $\{f^t(x_0, \mathbf{u})\} \cap \mathcal{X}_u = \emptyset$. This means that $\mathfrak{S}$ is robustly safe with respect to $\mathcal{X}_u$.

Proof of only if part. Now assume that the monotone dtDS $\mathfrak{S}$ is safe with respect to upper closed unsafe set $\mathcal{X}_u$, for every input $\{u_t\}_{t=0}^\infty \in U$. Let $\widehat{\Omega}$ be the lower closed set defined in (27) and $B : \mathcal{X} \rightarrow \mathbb{R}$ be the function defined in (28). Our goal is to show that $B : \mathcal{X} \rightarrow \mathbb{R}$ defined in (28) is a monotone K -inductive robust barrier certificate for $\mathfrak{S}$ satisfying conditions (13), (14), and (15). First, by Lemma D.3, the function $B : \mathcal{X} \rightarrow \mathbb{R}$ is monotone.

1. (**B satisfies condition (13)**) Suppose that $x \in \mathcal{X}_0$. By the definition of Ω , this implies that $f^k(x, \mathbf{u}) \in \Omega$, for every $k \in [K-1]$ and every $\mathbf{u} \in U^K$. As a result, we have $f^k(x, \mathbf{u}) \in \Omega \subseteq \widehat{\Omega}$, for every $k \in [K-1]$ and every $\mathbf{u} \in U^K$. Using the definition of B in (28), we get $B(f^k(x, \mathbf{u})) = \gamma$, for every $k \in [K-1]$ and every $\mathbf{u} \in U^K$. This shows that B satisfies condition (13).
2. (**B satisfies condition (14)**) Suppose that $x \in \mathcal{X}_u$. Since $\mathfrak{S}$ is safe with respect to $\mathcal{X}_u$ for any input $\{u_t\}_{t=0}^\infty \in U$, we get that $x \notin \Omega$. Note that $\widehat{\Omega}$ is the smallest lower closed set containing Ω and $\mathcal{X}_u$ is an upper closed set. As a result, $x \notin \widehat{\Omega}$. By definition of B in (28), we have $B(x) > \gamma$. This shows that B satisfies condition (14).
3. (**B satisfies condition (15)**) Let $x \in \mathcal{X}$ be such that $B(f^k(x, \mathbf{u})) \leq \gamma$, for every $k \in [K-1]$ and every $\mathbf{u} \in U^K$. By definition of B in (28), this implies that $f^k(x, \mathbf{u}) \in \widehat{\Omega}$, for every $k \in [K-1]$ and every $\mathbf{u} \in U^K$. Given an input $\mathbf{u} \in U^K$, by definition of the set $\widehat{\Omega}$ in (27), there exists $y \in \Omega$ such that $f^{K-1}(x, \mathbf{u}) \leq y$. Since the state transition map f is monotone, for every $u \in U$, we have $f(f^{K-1}(x, \mathbf{u}), u) \leq f(y, u)$. Note that $y \in \Omega$ and $u \in U$. Therefore, using Lemma D.2, we can show that $f(y, u) \in \Omega$. Using the definition of the lower closed set $\widehat{\Omega}$ in (27), this implies that $f(f^{K-1}(x, \mathbf{u}), u) \in \widehat{\Omega}$, for every $u \in U$. As a result, one can use the definition of B in (28) to show that $B(f(f^{K-1}(x, \mathbf{u}), u)) \leq \gamma$, for every $u \in U$. This shows that $B(f^K(x, \widehat{\mathbf{u}})) \leq \gamma$, for every $\widehat{\mathbf{u}} \in U^{K+1}$. Thus, B satisfies condition (13).

This completes the proof of Theorem 3.1 part 2.

E Proof of Theorem 3.2

Regarding part 1), suppose that $\mathcal{B}$ is a monotone function satisfying conditions (6)-(8). Our goal is to show that $\mathcal{B}$ is a monotone control K -inductive barrier certificate for $\mathfrak{S}$, i.e., it satisfies all the conditions (2)-(4).

1. (**$\mathcal{B}$ satisfies condition (2)**) Let $x \in \mathcal{X}_0$. Since $\mathcal{X}_0 \subseteq \cup_{i \in \mathcal{I}} [\underline{x}^i, \bar{x}^i]$, there exists $i \in \mathcal{I}$ such that $x \leq \bar{x}^i$. Since $\mathcal{B}$ satisfies (6), $\mathcal{B}(f^k(\bar{x}^i, \kappa(\bar{x}^i))) \leq \gamma$, $\forall k \in [K-1]$. By monotonicity of $\mathcal{B}$, the state-transition map f , and the control feedback κ , we get

$$\mathcal{B}(f^k(x, \kappa(x))) \leq \mathcal{B}(f^k(\bar{x}^i, \kappa(\bar{x}^i))) \leq \gamma, \quad \forall k \in [K-1].$$

2. (**$\mathcal{B}$ satisfies condition (3)**) Let $x \in \mathcal{X}_u$. There exists $j \in \mathcal{U}$ such that $\underline{x}^j \leq x$. Since $\mathcal{B}$ satisfies (7),

$$\eta \leq \mathcal{B}(\underline{x}^j) \leq \mathcal{B}(x).$$

3. (**$\mathcal{B}$ satisfies condition (4)**) Let $x \in \mathcal{X}$ be such that $\mathcal{B}(f^k(x, \kappa(x))) \leq \delta$, $\forall k \in [K-1]$. Choose $i \in \mathcal{P}$ such that $x \in [\underline{x}^i, \bar{x}^i]$. Using monotonicity of $\mathcal{B}$, the state-transition map f , and the control feedback κ , we get

$$\mathcal{B}(f^k(\underline{x}^i, \kappa(\underline{x}^i))) \leq \mathcal{B}(f^k(x, \kappa(x))) \leq \delta, \quad \forall k \in [K-1].$$

Since $\mathcal{B}$ satisfies (8), we get that $\mathcal{B}(f^K(\bar{x}^i, \kappa(\bar{x}^i))) \leq \delta$. This implies that,

$$\mathcal{B}(f^K(x, \kappa(x))) \leq \mathcal{B}(f^K(\bar{x}^i, \kappa(\bar{x}^i))) \leq \delta.$$

Therefore, $\mathcal{B}$ satisfies conditions (2)–(4), and hence is a K -inductive control barrier certificate for system $\mathfrak{S}$ with feedback controller κ . The conclusion then follows directly from Theorem 3.1, part 1, implying that the system $\mathfrak{S}$ is safely controllable with respect to $\mathcal{X}_u$.

Regarding part 2), suppose that $\mathcal{B}$ is a monotone function satisfying conditions (16)–(18). Our goal is to show that $\mathcal{B}$ is a monotone robust K -inductive barrier certificate for $\mathfrak{S}$, i.e., it satisfies all the conditions (13)–(15).

1. (**$\mathcal{B}$ satisfies condition (13)**) Let $x \in \mathcal{X}_0$. Since $\mathcal{X}_0 \subseteq \bigcup_{i \in \mathcal{I}} [\underline{x}^i, \bar{x}^i]$, there exists $i \in \mathcal{I}$ such that $x \leq \bar{x}^i$. Let $\mathbf{u} \in U^K$. Since $U \subseteq \bigcup_{j \in \mathcal{Q}} [\underline{u}^j, \bar{u}^j]$, there exists $\bar{\mathbf{u}} \in \left(\bigcup_{j \in \mathcal{Q}} \{\bar{u}^j\} \right)^K$ such that $\mathbf{u} \leq \bar{\mathbf{u}}$. Since $\mathcal{B}$ satisfies condition (16), we have

$$\mathcal{B}(f^k(\bar{x}^i, \bar{\mathbf{u}})) \leq \gamma, \quad \forall k \in [K-1].$$

On the other hand, $\mathcal{B}$ is monotone and f is a monotone state-transition map. Thus,

$$\mathcal{B}(f^k(x, \mathbf{u})) \leq \mathcal{B}(f^k(\bar{x}^i, \mathbf{u})) \leq \mathcal{B}(f^k(\bar{x}^i, \bar{\mathbf{u}})) \leq \gamma, \quad \forall k \in [K-1].$$

This shows that $\mathcal{B}$ satisfies condition (13).

2. (**$\mathcal{B}$ satisfies condition (14)**) Let $x \in \mathcal{X}_u$. Since $\mathcal{X}_u \subseteq \bigcup_{i \in \mathcal{U}} [\underline{x}^i, \bar{x}^i]$, there exists $j \in \mathcal{U}$ such that $\underline{x}^j \leq x$. Since $\mathcal{B}$ satisfies condition (17), we have $\eta \leq \mathcal{B}(\underline{x}^j)$. Using monotonicity of $\mathcal{B}$, we get

$$\eta \leq \mathcal{B}(\underline{x}^j) \leq \mathcal{B}(x).$$

Hence, $\mathcal{B}$ satisfies condition (14).

3. (**$\mathcal{B}$ satisfies condition (15)**) Let $x \in \mathcal{X}$ and $\mathbf{u} \in U^{K+1}$ be such that $\mathcal{B}(f^k(x, \mathbf{u})) \leq \delta, \forall k \in [K-1]$. Since the family of hyper-rectangles $\{[\underline{x}^i, \bar{x}^i]\}_{i \in \mathcal{P}}$ covers $\mathcal{X}$, there exists $i \in \mathcal{P}$ such that $x \in [\underline{x}^i, \bar{x}^i]$. Also, there exist

$$\bar{\mathbf{u}} \in \left(\bigcup_{j \in \mathcal{Q}} \{\bar{u}^j\} \right)^{K+1}, \quad \underline{\mathbf{u}} \in \left(\bigcup_{j \in \mathcal{Q}} \{\underline{u}^j\} \right)^{K+1},$$

such that $\underline{\mathbf{u}} \leq \mathbf{u} \leq \bar{\mathbf{u}}$. Using monotonicity of $\mathcal{B}$ and f , we get

$$\mathcal{B}(f^k(\underline{x}^i, \underline{\mathbf{u}})) \leq \mathcal{B}(f^k(x, \mathbf{u})) \leq \delta, \quad \forall k \in [K-1].$$

Since $\mathcal{B}$ satisfies condition (18), we have $\mathcal{B}(f^K(\bar{x}^i, \bar{\mathbf{u}})) \leq \delta$. As a result,

$$\mathcal{B}(f^K(x, \mathbf{u})) \leq \mathcal{B}(f^K(\bar{x}^i, \mathbf{u})) \leq \mathcal{B}(f^K(\bar{x}^i, \bar{\mathbf{u}})) \leq \delta,$$

where the first inequality holds by monotonicity of N and f together with $x \leq \bar{x}^i$, and the second inequality holds by monotonicity of $\mathcal{B}$ and f together with $\mathbf{u} \leq \bar{\mathbf{u}}$. Hence, $\mathcal{B}$ satisfies condition (15).

Therefore, $\mathcal{B}$ satisfies conditions (13)–(15). Hence, $\mathcal{B}$ is a K -inductive robust barrier certificate. The result then follows from Theorem 3.1 part 2. Therefore, the system $\mathfrak{S}$ is robustly safe with respect to $\mathcal{X}_u$. This completes the proof.

F Memory-efficient inductive barrier certificates

In this appendix, by leveraging the monotonicity of the barrier certificate, we propose the following memory-efficient formulation of control barrier certificates. Let $\gamma, \delta, \eta \in \mathbb{R}$ satisfy $\gamma \leq \delta < \eta$. We have:

$$\mathcal{B}(f(\bar{x}^i, \kappa(\bar{x}^i))) \geq \mathcal{B}(\bar{x}^i) \quad \forall i \in \mathcal{I}, \quad (29)$$

$$\mathcal{B}(f(\underline{x}^i, \kappa(\underline{x}^i))) \geq \mathcal{B}(\underline{x}^i) \quad \forall i \in \mathcal{P}, \quad (30)$$

$$\mathcal{B}(f^k(\bar{x}^i, \kappa(\bar{x}^i))) \leq \gamma \quad \forall i \in \mathcal{I}, \quad (31)$$

$$\mathcal{B}(\underline{x}^i) \geq \eta \quad \forall i \in \mathcal{U}, \quad (32)$$

$$\mathcal{B}(f^k(\underline{x}^i, \kappa(\underline{x}^i))) \leq \delta \Rightarrow \mathcal{B}(f^{k+1}(\bar{x}^i, \kappa(\bar{x}^i))) \leq \delta, \quad \forall i \in \mathcal{P} \quad (33)$$

Similarly, we can propose the following memory-efficient formulation of robust barrier certificates:

$$\mathcal{B}(f(\bar{x}^i, \bar{u}^j)) \geq \mathcal{B}(\bar{x}^i) \quad \forall i \in \mathcal{I}, \forall j \in \mathcal{Q}, \quad (34)$$

$$\mathcal{B}(f(\underline{x}^i, \underline{u}^j)) \geq \mathcal{B}(\underline{x}^i) \quad \forall i \in \mathcal{P}, \forall j \in \mathcal{Q}, \quad (35)$$

$$\mathcal{B}(f^k(\bar{x}^i, \bar{\mathbf{u}})) \leq \gamma \quad \forall i \in \mathcal{I}, \quad (36)$$

$$\mathcal{B}(\underline{x}^i) \geq \eta \quad \forall i \in \mathcal{U}, \quad (37)$$

$$\mathcal{B}(f^k(\underline{x}^i, \underline{\mathbf{u}})) \leq \delta \Rightarrow \mathcal{B}(f^{k+1}(\bar{x}^i, \bar{\mathbf{u}})) \leq \delta, \quad \forall i \in \mathcal{P} \quad (38)$$

The following results demonstrate the effectiveness of the memory-efficient formulation of control and robust barrier certificates.

Corollary F.1 (Memory-efficient inductive barrier certificates). *Consider a monotone dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ with an unknown state-transition map f and a set of unsafe states $\mathcal{X}_u$. Let $\{[\underline{x}^i, \bar{x}^i]\}_{i \in \mathcal{P}}$ be a family of hyper-rectangles covering the state space $\mathcal{X}$, and let $\{[\underline{u}^j, \bar{u}^j]\}_{j \in \mathcal{Q}}$ be a family of hyper-rectangles covering the input space U . Define the index sets $\mathcal{I}$ and $\mathcal{U}$ as the minimal subsets of $\mathcal{P}$ satisfying (5). Then the system $\mathfrak{S}$*

1. (Safe Control Synthesis) *is safely controllable with respect to $\mathcal{X}_u$ if there exist monotone inductive barrier certificate $\mathcal{B} : \mathcal{X} \rightarrow \mathbb{R}$ and a monotone feedback $\kappa : \mathcal{X} \rightarrow U$ satisfying conditions (29)–(33).*
2. (Robust Safety Verification) *is robustly safe with respect to $\mathcal{X}_u$ if there exists a monotone inductive barrier certificate $\mathcal{B} : \mathcal{X} \rightarrow \mathbb{R}$ satisfying conditions (34)–(38).*

Corollary F.1 enables safe controller synthesis and verification of safety with reduced memory requirements and improved computational efficiency. Therefore, throughout this paper, we can replace 16–18 with 34–38, or 6–8 with 29–33, to obtain a memory-efficient version of the robust barrier or safe controller.

G Inductive vs. classic monotone barriers

We consider a discrete-time freeway traffic system [13] $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$ with state $q(t) = [q_1(t), q_2(t)]^\top$, where $q_i(t)$ denotes the vehicle density in road segment i at time t . The state space is $\mathcal{X} = [0, 50] \times [0, 50]$. We fix the sampling time to $\tau = 0.5$ and the upstream inflow to $F_{0 \rightarrow 1} = 8.3$. The flow from segment 1 to segment 2 is given by the demand–supply rule $F_{1 \rightarrow 2}(q(t)) = \min\{\varphi(q_1(t)), \sigma(q_2(t))\}$, where the demand and supply functions are defined as $\varphi(x) = 10(1 - e^{-0.1x})$ and $\sigma(x) = \max\{0, 10(1 - (x/50)^3)\}$. The resulting discrete-time dynamics are

$$q_1(t+1) = q_1(t) + \tau(F_{0 \rightarrow 1} - F_{1 \rightarrow 2}(q(t))), \quad q_2(t+1) = q_2(t) + \tau(F_{1 \rightarrow 2}(q(t)) - \varphi(q_2(t))),$$

followed by component-wise saturation to enforce $q(t+1) \in \mathcal{X}$. The initial set is $\mathcal{X}_0 = [0, 10] \times [0, 10]$, and the unsafe set is defined as $\mathcal{X}_u = \mathcal{X} \setminus ([0, 40] \times [0, 40])$, which captures congestion whenever at least one segment density exceeds 40.

To verify safety using only sampled data, we employ a fixed hyper-rectangular gridding of the state space. Specifically, we partition the unsafe region into two rectangles, namely $[0, 40] \times [40, 50]$ and $[40, 50] \times [0, 50]$, and define one rectangle for the initial set $[0, 10] \times [0, 10]$, together with a larger rectangle $[0, 40] \times [0, 40]$ covering the remaining safe domain. This results in a total of four hyper-rectangles and five representative corner points $\{\bar{x}_{in}, \underline{x}_{us}^1, \underline{x}_{us}^2, \underline{x}_s, \bar{x}_s\}$. For the corner point $\bar{x}_s$, we generate trajectory samples by evaluating the one-step transition $f(\bar{x}_s)$. Figure 1 illustrates the resulting rectangular gridding, where the sampled points $\{\bar{x}_{in}, \underline{x}_{us}^1, \underline{x}_{us}^2, \underline{x}_s, \bar{x}_s\}$ are marked with squares and $f(\bar{x}_s)$ is marked with triangle.

First, we train a neural inductive barrier certificate using the sampled data collected from this grid and verify safety via Algorithm J.1. Using a simplified Max-Monotone Network based on the direct max-input branch in Definition 4.1, with a monotone hidden layer of 5 neurons and $K = 100$, our method successfully verifies the safety of the 2-D traffic system in 0.05 s. Figure 2 illustrates the resulting sub-level set of the learned neural inductive barrier over the state space. In contrast, a classic ($K = 1$) monotone neural barrier certificate cannot certify safety under the same fixed gridding. In

particular, although $B(\underline{x}_s) \leq 0$ holds since $\underline{x}_s \in \mathcal{X}_0$, we have $B(f(\bar{x}_s)) > 0$, which violates the barrier condition. Consequently, safety cannot be verified using an ordinary monotone barrier on this coarse partition.

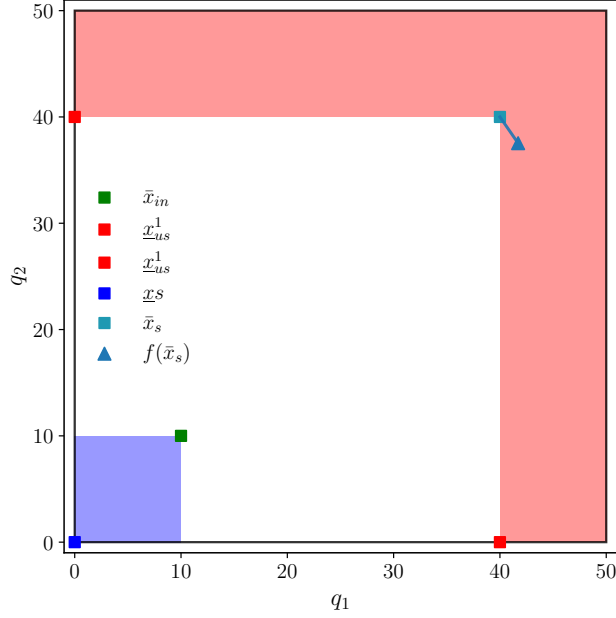


Figure 1: Extreme-point evaluation of one-step dynamics.

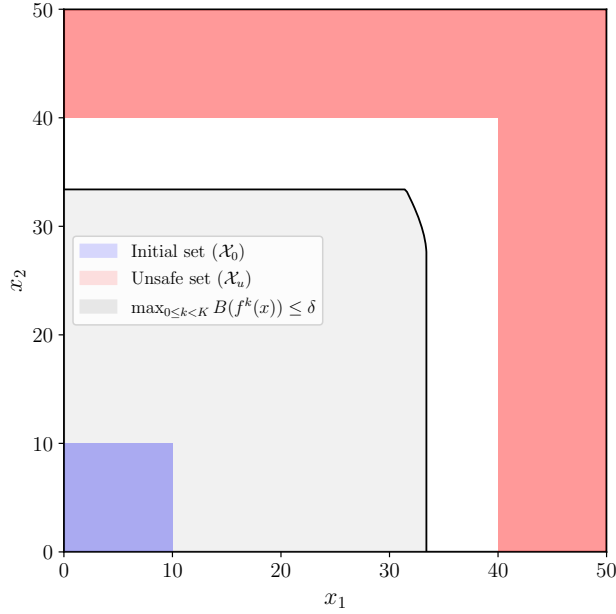


Figure 2: Sub-level set of the learned neural inductive barrier certificate over the state space. The parameters $\gamma = 5$, $\delta = 9.5$, and $\eta = 10$ are used in this experiment.

H Max-Monotone Network

In this appendix, we present additional theoretical results and technical details concerning the Max-Monotone Networks introduced in Definition 4.1.

H.1 Max-separable inductive barrier certificates

In this appendix, we show that, under mild assumptions, the existence of a monotone inductive barrier certificate for a monotone system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ implies the existence of a monotone max-separable² inductive barrier certificate for $\mathfrak{S}$.

Theorem H.1 (Max-separable monotone inductive barrier certificates). *Consider a monotone dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ with compact state space $\mathcal{X} \subseteq \mathbb{R}^N$, and a compact unsafe set $\mathcal{X}_u \subseteq \mathcal{X}$, and $K \in \mathbb{N}_{\geq 1}$. Suppose that there exists a Lipschitz monotone control K -inductive barrier certificate $\mathcal{B}$ for system $\mathfrak{S}$ satisfying conditions (2)–(4) for some $\gamma < \delta < \eta$. Then, there exists a max-separable monotone control K -inductive barrier certificate for $\mathfrak{S}$ satisfying conditions (2)–(4).*

Proof. For simplicity of notation, we present the proof for the case $K = 1$. The proof for $K > 1$ follows analogously, but involves substantially heavier notation that we omit for clarity. Since $\mathcal{B}$ is a monotone control K -inductive barrier certificate for $\mathfrak{S}$, we get

$$\mathcal{B}(x) \leq \gamma \quad \forall x \in \mathcal{X}_0, \quad (39)$$

$$\mathcal{B}(x) \geq \eta \quad \forall x \in \mathcal{X}_u, \quad (40)$$

$$\mathcal{B}(x) \leq \delta \implies \exists u \in U \text{ s.t. } \mathcal{B}(f(x, u)) \leq \delta \quad \forall x \in \mathcal{X}. \quad (41)$$

Since $\mathcal{B}$ is a Lipschitz function on the compact domain $\mathcal{X}$, there exists $L_{\mathcal{B}} > 0$ such that

$$|\mathcal{B}(x) - \mathcal{B}(y)| \leq L_{\mathcal{B}} \|x - y\|_{\infty}, \quad \forall x, y \in \mathcal{X}.$$

Additionally, since $\mathcal{B}$ is continuous and $\mathcal{X}$ is compact, there exist constant m such that $m = \min_{x \in \mathcal{X}} \mathcal{B}(x)$. On the other hand, continuity of $\mathcal{B}$ implies that the set $\mathcal{B}_{\leq \delta} := \{x \in \mathcal{X} \mid \mathcal{B}(x) \leq \delta\}$ is closed in $\mathcal{X}$ and thus compact. Using the fact that $\delta < \eta$, we get

$$\mathcal{B}_{\leq \delta} \cap \mathcal{X}_u = \emptyset.$$

Since both sets $\mathcal{B}_{\leq \delta}$ and $\mathcal{X}_u$ are compact, there exists $\epsilon > 0$ such that:

$$(\mathcal{B}_{\leq \delta} \oplus S_{\epsilon}) \cap (\mathcal{X}_u \oplus S_{\epsilon}) = \emptyset. \quad (42)$$

where $\oplus$ is the Minkowski sum and $S_{\epsilon} = \{x \in \mathbb{R}^n \mid \|x\|_{\infty} \leq \epsilon\}$ is the ℓ_{∞} -norm ball of radius ϵ . Without loss of generality, we can pick ϵ small enough such that $L_{\mathcal{B}}\epsilon \leq |\gamma - \delta|$. Note that the state set $\mathcal{X}$ is compact. Therefore, there exists an ℓ_{∞} -norm ball $S_R = \{x \in \mathbb{R}^n \mid \|x\|_{\infty} \leq R\}$ such that $\mathcal{X} \subseteq S_R$ for some large enough $R > 0$. We consider the following uniform hyper-rectangular cover of $\mathcal{X}$:

$$\{[\mathbf{v}\epsilon, (\mathbf{v} + \mathbf{1}_n)\epsilon]\}_{\mathbf{v} \in \mathbb{Z}_{\ell}^n} \quad (43)$$

where $\mathbb{Z}_{\ell} = \{-\ell, -\ell + 1, \dots, 0, \dots, \ell - 1, \ell\}$ and $\ell = \lceil \frac{R}{\epsilon} \rceil$. We define the set $\mathcal{A} \subseteq \mathbb{R}^n$ by

$$\mathcal{A} := \{(\mathbf{v} + \mathbf{1}_n)\epsilon \mid \mathcal{B}_{\leq (\delta - L_{\mathcal{B}}\epsilon)} \cap [\mathbf{v}\epsilon, (\mathbf{v} + \mathbf{1}_n)\epsilon] \neq \emptyset\}.$$

Moreover, we define the function $N : \mathcal{X} \rightarrow \mathbb{R}$ by

$$N(x) = \begin{cases} m & x \leq z \text{ for some } z \in \mathcal{A}, \\ \eta & \text{otherwise.} \end{cases} \quad (44)$$

It is easy to check that $N(x) \leq \mathcal{B}(x)$, for every $x \in \mathcal{X}$. We show that N in (44) is a monotone max-separable function and it satisfies conditions (39)–(41).

1. (**N satisfies condition (39)**) Let $x \in \mathcal{X}_0$. Then we have $\mathcal{B}(x) \leq \gamma \leq \delta - L_{\mathcal{B}}\epsilon$. This means that $x \in \mathcal{B}_{\leq (\delta - L_{\mathcal{B}}\epsilon)}$. Since the partition (43) covers the whole state set $\mathcal{X}$, there exists $\mathbf{v} \in \mathbb{Z}_{\ell}^n$ such that $x \in [\mathbf{v}\epsilon, (\mathbf{v} + \mathbf{1}_n)\epsilon]$. This implies that $x \in \mathcal{B}_{\leq (\delta - L_{\mathcal{B}}\epsilon)} \cap [\mathbf{v}\epsilon, (\mathbf{v} + \mathbf{1}_n)\epsilon]$. As a result $(\mathbf{v} + \mathbf{1}_n)\epsilon \in \mathcal{A}$. Since $x \leq (\mathbf{v} + \mathbf{1}_n)\epsilon$, by definition of N in (44) we get $N(x) = m \leq \gamma$.

²A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is max-separable, if there exist scalar functions $f_i : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = \max_{i \in \{1, \dots, n\}} f_i(x_i)$.

2. (**N satisfies condition (40)**) Let $x \in \mathcal{X}_u$. Since the partition (43) covers the whole state set $\mathcal{X}$, there exists $\mathbf{v} \in \mathbb{Z}_\ell^n$ such that $x \in [\mathbf{v}\epsilon, (\mathbf{v} + \mathbf{1}_n)\epsilon]$. Our goal is to show that $(\mathbf{v} + \mathbf{1}_n)\epsilon \notin \mathcal{A}$. We prove this by contradiction. Assume that $(\mathbf{v} + \mathbf{1}_n)\epsilon \in \mathcal{A}$. This means that there exists $y \in \mathcal{B}_{\leq(\delta-L_B\epsilon)} \subseteq \mathcal{B}_{\leq\delta}$ such that $y \in [\mathbf{v}\epsilon, (\mathbf{v} + \mathbf{1}_n)\epsilon]$. As a result, we get $\|y - x\|_\infty \leq \epsilon$. However, this is not possible because $x \in \mathcal{X}_u$ and $y \in \mathcal{B}_{\leq\delta}$ and the intersection in (42) is empty. Therefore, $(\mathbf{v} + \mathbf{1}_n)\epsilon \notin \mathcal{A}$ and therefore by definition $N(x) = \eta$.
3. (**N satisfies condition (41)**) Let $x \in \mathcal{X}$ be such that $N(x) \leq \delta$. By definition of N in (44), this implies that there exists $\mathbf{v}$ such that $(\mathbf{v} + \mathbf{1}_n)\epsilon \in \mathcal{A}$ and $x \leq (\mathbf{v} + \mathbf{1}_n)\epsilon$. As a result, there exists $y \in \mathcal{B}_{\leq(\delta-L_B\epsilon)}$ such that $y \in [\mathbf{v}\epsilon, (\mathbf{v} + \mathbf{1}_n)\epsilon]$. Using the Lipschitz property of $\mathcal{B}$, we get

$$\mathcal{B}(x) \leq \mathcal{B}(y) + L_B\|x - y\|_\infty \leq \mathcal{B}(y) + L_B\epsilon \leq \delta.$$

This implies that $\mathcal{B}(x) \leq \delta$. Since $\mathcal{B}$ satisfies condition (41), there exists $u \in U$ such that $\mathcal{B}(f(x, u)) \leq \delta$. Since $N(w) \leq \mathcal{B}(w)$, for every $w \in \mathcal{X}$, we get $N(f(x, u)) \leq \mathcal{B}(f(x, u)) \leq \delta$.

4. (**N is monotone and max-separable**) Let $x \leq y$. We consider two cases: (i) $y \leq z$ for some $z \in \mathcal{A}$, and (ii) $y \not\leq z$ for every $z \in \mathcal{A}$. In case (i), by definition of N in (44), we have $N(y) = m$. Moreover, since $x \leq y \leq z$, we also get $N(x) = m$. Therefore, $N(x) = N(y)$. In case (ii), by definition of N in (44), we have $N(y) = \eta$. Since $N(w) \leq \eta$ for every $w \in \mathcal{X}$, we get $N(x) \leq N(y)$. This completes the proof of monotonicity of N .

Now, consider the scalar functions $N_i : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$N_i(s) = \begin{cases} m & s \leq z_i \text{ for some } z = (z_1, \dots, z_n) \in \mathcal{A}, \\ \eta & \text{otherwise.} \end{cases} \quad (45)$$

Our goal is to show that $N(x) = \max_{i \in \{1, \dots, n\}} N_i(x_i)$, for every $x \in \mathcal{X}$. We consider two cases: (i) $x \leq z$ for some $z \in \mathcal{A}$, and (ii) $x \not\leq z$ for every $z \in \mathcal{A}$.

In case (i), by definition of N in (44), we get $N(x) = m$. On the other hand, $x_i \leq z_i$, for every $i \in \{1, \dots, n\}$. Thus, by definition of N_i in (45), we have $N_i(x_i) = m$. This means that

$$N(x) = m = \max_{i \in \{1, \dots, n\}} N_i(x_i).$$

In case (ii), for every $z \in \mathcal{A}$ there exists $i \in \{1, \dots, n\}$ such that $x_i > z_i$. Hence, by definition of N_i in (45), there exists $i \in \{1, \dots, n\}$ such that $N_i(x_i) = \eta$. Since $N_j(x_j) \leq \eta$ for every $j \in \{1, \dots, n\}$, we get

$$\max_{i \in \{1, \dots, n\}} N_i(x_i) = \eta = N(x).$$

Therefore, N is max-separable. □

H.2 Architecture of Max-Monotone Networks

Figure 3 provides a schematic illustration of the proposed Max-Monotone Network architecture defined in Section 4.1. The network consists of a direct max-input branch and multiple affine max branches, each followed by a monotone sub-network, with the final output obtained via a minimum operator.

H.3 Proof of Theorem 4.2

We first establish monotonicity, and then prove the universal approximation property.

Monotonicity part. Let $x, y \in \mathbb{R}^n$ such that $x \leq y$ componentwise. We show that $N(x) \leq N(y)$.

First, for $j = 0$, we have

$$g^{(0)}(x) = \max_i x_i \leq \max_i y_i = g^{(0)}(y).$$

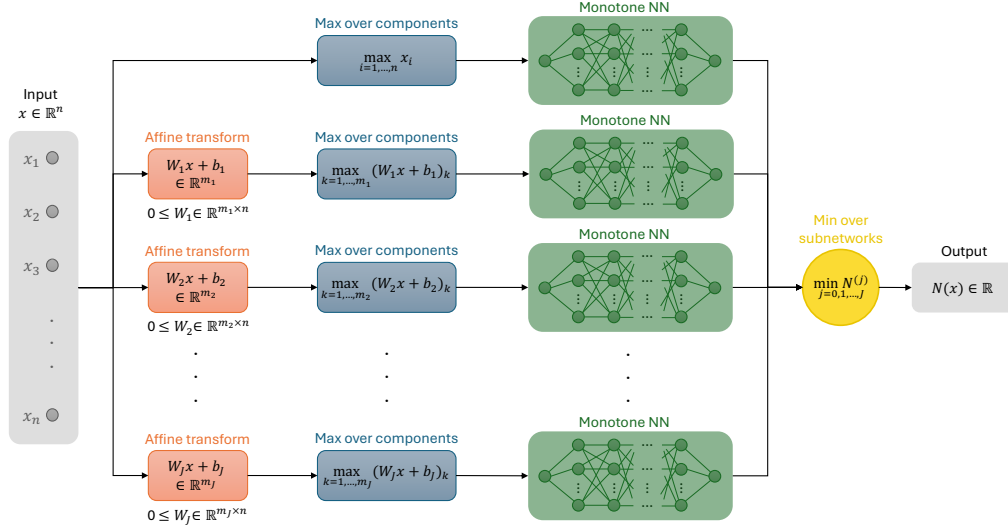


Figure 3: Schematic illustration of the proposed Max-Monotone Network architecture.

For $j = 1, \dots, J$, since $W_j \geq 0$, it follows that $W_j x \leq W_j y$ componentwise, and hence

$$(W_j x + b_j)_k \leq (W_j y + b_j)_k, \quad \forall k.$$

Taking the maximum over k yields

$$g^{(j)}(x) \leq g^{(j)}(y), \quad \forall j = 1, \dots, J.$$

Since each $N^{(j)}$ is a monotone function, we obtain

$$N^{(j)}(g^{(j)}(x)) \leq N^{(j)}(g^{(j)}(y)), \quad \forall j = 0, \dots, J.$$

Taking the minimum over j preserves the inequality, and therefore

$$N(x) = \min_j N^{(j)}(g^{(j)}(x)) \leq \min_j N^{(j)}(g^{(j)}(y)) = N(y).$$

Thus, N is monotone.

Universal approximation part. Let $m : [0, 1]^n \rightarrow \mathbb{R}$ be continuous, bounded, and monotone with bounded partial derivatives. Fix $\varepsilon > 0$. Let $a = \min_{x \in [0, 1]^n} m(x)$, $b = \max_{x \in [0, 1]^n} m(x)$, and let $\alpha > 0$ be a bound on the magnitude of all partial derivatives of m . We first note that the direct-max branch $N^{(0)}(g^{(0)}(x))$ does not affect the approximation argument. Indeed, choose $N^{(0)}$ to be a constant function satisfying

$$N^{(0)}(g^{(0)}(x)) > b, \quad \forall x \in [0, 1]^n.$$

Hence, this branch is never selected by the final minimum. It therefore suffices to construct the approximation using only the max-affine branches $j = 1, \dots, J$. Partition $[0, 1]^n$ into a uniform grid with spacing $\delta = \frac{1}{N}$ along each dimension, and denote the set of grid points by

$$S = \{(i_1 \delta, \dots, i_n \delta) : 1 \leq i_d \leq N\}.$$

For each grid point $x' \in S$, we construct one branch indexed by j and define

$$g^{(j)}(x) = \max_{k=1, \dots, n+1} ((W_j x + b_j)_k),$$

where the affine components are chosen as follows: one component is the constant $m(x')$, for each coordinate $d = 1, \dots, n$, one component is

$$m(x') + \gamma(x_d - x'_d),$$

with $\gamma > \frac{b-a}{\delta}$. This construction is of the form $g^{(j)}(x) = \max_k((W_j x + b_j)_k)$ with $W_j \geq 0$. For each such branch, choose the scalar monotone neural network $N^{(j)}$ to be the identity map, i.e.,

$$N^{(j)}(t) = t.$$

This is realized exactly by a monotone neural network with a single linear layer of weight 1 and bias 0. Hence, for all $j \geq 1$,

$$N^{(j)}(g^{(j)}(x)) = g^{(j)}(x).$$

We now show that this construction approximates m . Fix $x \in [0, 1]^n$, and let $s^{(1)} \in S$ be the largest grid point such that $s^{(1)} \leq x$ componentwise, i.e.,

$$0 \leq x_d - s_d^{(1)} < \delta, \quad \forall d.$$

Define $s^{(2)} = s^{(1)} + (\delta, \dots, \delta)$. Consider the branch associated with $s^{(2)}$. Since $x_d - s_d^{(2)} \leq 0$ for all d , every affine component in that branch satisfies

$$m(s^{(2)}) + \gamma(x_d - s_d^{(2)}) \leq m(s^{(2)}).$$

Hence, the maximum is attained at the constant component, and therefore

$$g^{(j)}(x) = m(s^{(2)}).$$

Since $N^{(j)}$ is the identity and $N(x) = \min_j N^{(j)}(g^{(j)}(x))$, we obtain

$$N(x) \leq N^{(j)}(g^{(j)}(x)) = g^{(j)}(x) = m(s^{(2)}).$$

Next, consider any branch associated with a grid point $s' \in S$. If s' dominates $s^{(1)}$, then by monotonicity of m ,

$$m(s') \geq m(s^{(1)}).$$

Since $m(s')$ is one of the affine components in that branch, it follows that

$$g^{(j)}(x) \geq m(s^{(1)}).$$

On the other hand, if s' does not dominate $s^{(1)}$, then there exists a coordinate d such that

$$s'_d < s_d^{(1)}.$$

Since $0 \leq x_d - s_d^{(1)} < \delta$, this implies

$$x_d - s'_d \geq \delta.$$

Hence, the corresponding affine component satisfies

$$m(s') + \gamma(x_d - s'_d) \geq a + \gamma\delta > a + (b - a) = b \geq m(s^{(1)}).$$

Therefore, every branch output is at least $m(s^{(1)})$, and thus

$$N(x) \geq m(s^{(1)}).$$

Combining the two inequalities gives

$$m(s^{(1)}) \leq N(x) \leq m(s^{(2)}).$$

Since m is monotone and $s^{(1)} \leq x \leq s^{(2)}$, we also have

$$m(s^{(1)}) \leq m(x) \leq m(s^{(2)}).$$

Therefore,

$$|m(x) - N(x)| \leq |m(s^{(2)}) - m(s^{(1)})|.$$

By the mean value theorem, there exists a point c on the line segment joining $s^{(1)}$ and $s^{(2)}$ such that

$$m(s^{(2)}) - m(s^{(1)}) = \sum_{d=1}^n \frac{\partial m(c)}{\partial x_d} (s_d^{(2)} - s_d^{(1)}).$$

Since $s_d^{(2)} - s_d^{(1)} = \delta$ for all d , and $|\partial m / \partial x_d| \leq \alpha$, it follows that

$$|m(s^{(2)}) - m(s^{(1)})| \leq \sum_{d=1}^n \left| \frac{\partial m(c)}{\partial x_d} \right| \delta \leq n\delta\alpha.$$

Choosing $\delta < \frac{\varepsilon}{n\alpha}$ yields

$$|m(x) - N(x)| < \varepsilon, \quad \forall x \in [0, 1]^n.$$

This completes the proof.

I Safety certificate form zero loss

In the next theorem, we show that the loss functions defined in (10)–(12) not only promote the learned monotone neural network to satisfy conditions (6)–(8), but also guarantee that, whenever these loss functions vanish, the resulting network constitutes an inductive barrier certificate for $\mathfrak{S}$.

Theorem I.1 (Zero-loss certification). *Consider a monotone dtDS $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ with an unknown state-transition map f and a set of unsafe states $\mathcal{X}_u$. Let $\{\underline{x}^i, \bar{x}^i\}_{i \in \mathcal{P}}$ be a family of hyper-rectangles covering the state space $\mathcal{X}$, and let $\{\underline{u}^j, \bar{u}^j\}_{j \in \mathcal{Q}}$ be a family of hyper-rectangles covering the input space U . Define the index sets $\mathcal{I}$ and $\mathcal{U}$ as the minimal subsets of $\mathcal{P}$ satisfying (5). Assume $\gamma < \delta < \eta$. Then,*

- 1) (Safe control synthesis) *If there exist monotone neural networks $N : \mathcal{X} \rightarrow \mathbb{R}$ and $\kappa : \mathcal{X} \rightarrow U$ such that the loss in (10)–(12) satisfies $L_{\text{total}}^s = 0$, then the system $\mathfrak{S}$ is safely controllable with respect to $\mathcal{X}_u$ with the feedback controller κ .*
- 2) *If there exists a monotone neural network $N : \mathcal{X} \rightarrow \mathbb{R}$ such that the loss in (19)–(21) satisfies $L_{\text{total}}^v = 0$, then the system $\mathfrak{S}$ is robustly safe with respect to $\mathcal{X}_u$.*

Proof. Regarding part 1), our goal is to show that if there exist monotone neural networks $N : \mathcal{X} \rightarrow \mathbb{R}$ and $\kappa : \mathcal{X} \rightarrow U$ such that $L_{\text{total}}^s = L_1^s + L_2^s + L_3^s = 0$, then the system $\mathfrak{S}$ is safely controllable with respect to $\mathcal{X}_u$. Since each loss term is nonnegative, we obtain $L_1^s = L_2^s = L_3^s = 0$. We now show that (N, κ) satisfies conditions (6)–(8).

1. $((N, \kappa)$ **satisfies condition (6)**) Since $L_1^s = 0$, every summand in (10) is zero. Hence, for every $i \in \mathcal{I}$,

$$\max_{k \in [K-1]} N(f^k(\bar{x}^i, \kappa(\bar{x}^i))) \leq \gamma,$$

which gives (6).

2. $(N$ **satisfies condition (7)**) Since $L_2^s = 0$, every summand in (11) is zero. Hence, for every $i \in \mathcal{U}$,

$$N(\underline{x}^i) \geq \eta,$$

which gives (7).

3. $((N, \kappa)$ **satisfies condition (8)**) Since $L_3^s = 0$, every summand in (12) is zero. Therefore, whenever

$$\max_{k \in [K-1]} N(f^k(\underline{x}^i, \kappa(\underline{x}^i))) \leq \delta,$$

we have

$$N(f^K(\bar{x}^i, \kappa(\bar{x}^i))) \leq \delta,$$

which gives (8).

This implies that, (N, κ) satisfies conditions (6)–(8). Now we can use Theorem 3.2 part 1) to deduce that the system $\mathfrak{S}$ is safely controllable with respect to $\mathcal{X}_u$.

Regarding part 2), our goal is to show that if there exists a monotone neural network $N : \mathcal{X} \rightarrow \mathbb{R}$ such that

$$L_{\text{total}}^v = L_1^v + L_2^v + L_3^v = 0,$$

then the system $\mathfrak{S}$ is robustly safe with respect to $\mathcal{X}_u$. Since each loss term is nonnegative, we obtain

$$L_1^v = L_2^v = L_3^v = 0.$$

We first show that N satisfies conditions (16)–(18).

1. $(N$ **satisfies condition (16)**) Since $L_1^v = 0$, every summand in (19) must be zero. Hence, for every $i \in \mathcal{I}$ and $j \in \mathcal{Q}$,

$$\text{ReLU}\left(\max_{k \in [K-1]} (N(f^k(\bar{x}^i, \bar{u}^j)) - \gamma)\right) = 0.$$

Therefore, $\max_{k \in [K-1]} N(f^k(\bar{x}^i, \bar{u}^j)) \leq \gamma$. Hence, condition (16) holds.

2. (**N satisfies condition (17)**) Since $L_2^y = 0$, every summand in (20) must be zero. Hence, for every $i \in \mathcal{U}$,

$$\text{ReLU}(\eta - N(\underline{x}^i)) = 0.$$

Thus, $N(\underline{x}^i) \geq \eta$. Hence, condition (17) holds.

3. (**N satisfies condition (18)**) Since $L_3^y = 0$, every summand in (21) must be zero. Let $i \in \mathcal{P}$ and $j \in \mathcal{Q}$. If

$$\max_{k \in [K-1]} N(f^k(\underline{x}^i, \underline{u}^j)) \leq \delta,$$

then by the definition of r_{ij} we have $r_{ij} = 1$. Therefore, $\text{ReLU}(N(f^K(\bar{x}^i, \bar{u}^j)) - \delta) = 0$, which implies

$$N(f^K(\bar{x}^i, \bar{u}^j)) \leq \delta.$$

Hence, condition (18) holds.

This implies that, N satisfies conditions (16)–(18). Now we can use Theorem 3.2 part 2) to deduce that the system $\mathfrak{S}$ is robustly safe with respect to $\mathcal{X}_u$. This completes the proof. $\square$

J Algorithms

J.1 Monotone Neural Robust Barrier Certificate

Algorithm 2 Monotone neural robust barrier learning with adaptive refinement

Inputs: Simulator access to the dynamics f ; hyper-rectangular covers $\{[\underline{x}^i, \bar{x}^i]\}_{i \in \mathcal{P}}$, $\{[\underline{x}_0^i, \bar{x}_0^i]\}_{i \in \mathcal{I}}$, and $\{[\underline{x}_u^j, \bar{x}_u^j]\}_{j \in \mathcal{U}}$, of $\mathcal{X}$, $\mathcal{X}_0$, and $\mathcal{X}_u$, respectively; an input cover $\{[\underline{u}^q, \bar{u}^q]\}_{q \in \mathcal{Q}}$ of U ; margins $\gamma < \delta < \eta$; maximum state-refinement budget $n_{\max}^{\mathcal{P}}$; maximum input-refinement budget $n_{\max}^{\mathcal{Q}}$; maximum inductive parameter $K_{\max}$; maximum number of epochs N_{ep} ; and constants $\epsilon > 0$ and $C_{\max} > 1$.

Output: A monotone neural robust barrier certificate N and final loss L_{total}^v , if found; otherwise FAIL and final loss L_{total}^v .

```

1: Initialize  $\mathcal{H} \leftarrow \{[\underline{x}^\ell, \bar{x}^\ell]\}_{\ell \in \mathcal{P} \setminus \mathcal{U}}$ ,  $\mathcal{J} \leftarrow \{[\underline{u}^q, \bar{u}^q]\}_{q \in \mathcal{Q}}$ ,  $\mathcal{C} \leftarrow \emptyset$ ,  $L_{\text{total}}^v \leftarrow \infty$ .
2: for  $p = 1, \dots, n_{\max}^{\mathcal{P}}$  do
3:   for  $K = 1, \dots, K_{\max}$  do
4:     Define  $L_{\text{shrink}} = \sum_{x^c \in \mathcal{C}} [\text{ReLU}(\eta - N(x^c))]^2$ .
5:     Train  $N$  for at most  $N_{\text{ep}}$  epochs using  $L_{\text{total}}^v = L_1^v + L_2^v + L_3^v + L_{\text{shrink}}$ , where
        $L_1^v, L_2^v, L_3^v$  are defined in (19)–(21) using  $\mathcal{I}, \mathcal{U}$ , the current input-cover index set induced
       by  $\mathcal{J}$ , the current unresolved index set induced by  $\mathcal{H}$ , the current value of  $K$ , and margins
        $\gamma, \delta, \eta$ .
6:     if  $L_{\text{total}}^v = 0$  then
7:       return  $N, L_{\text{total}}^v$ 
8:     end if
9:   end for
10:  Apply either Algorithm 3 or Algorithm 4 to obtain refined state collections  $(\mathcal{H}^+, \mathcal{C}^+)$ 
11:  if the chosen state-refinement algorithm returns FAIL then
12:    return FAIL,  $L_{\text{total}}^v$ 
13:  else
14:    Update  $(\mathcal{H}, \mathcal{C})$  using the returned  $(\mathcal{H}^+, \mathcal{C}^+)$ 
15:  end if
16:  if  $|\mathcal{J}| < n_{\max}^{\mathcal{Q}}$  then
17:    Apply Algorithm 5 to obtain a refined input cover  $\mathcal{J}^+$ 
18:    Update  $\mathcal{J} \leftarrow \mathcal{J}^+$ 
19:  end if
20: end for
21: return FAIL,  $L_{\text{total}}^v$ 

```

J.2 Monotone Neural Control Barrier Certificate

Algorithm 1 presents the control barrier learning procedure.

J.3 Monotone-Shrinking Hyper-Rectangular Search

Algorithm 3 Monotone shrinking of hyper-rectangular grid

Inputs: A trained monotone neural certificate N ; the current unresolved hyper-rectangular collection $\mathcal{H} = \{[\underline{x}^\ell, \bar{x}^\ell]\}_{\ell \in \mathcal{P} \setminus \mathcal{U}}$; the current shrinking-constraint set $\mathcal{C}$.

Output: A refined unresolved collection $\mathcal{H}^+$ and an updated shrinking-constraint set $\mathcal{C}^+$; or FAIL.

```

1: Initialize  $\mathcal{H}^+ \leftarrow \emptyset$  and  $\mathcal{C}^+ \leftarrow \mathcal{C}$ 
2: for each box  $[\underline{x}^\ell, \bar{x}^\ell] \in \mathcal{H}$  do
3:   for  $i = 1, \dots, n$  do
4:     Attempt to find  $\xi_i^\ell \in [\underline{x}_i^\ell, \bar{x}_i^\ell]$  such that
        
$$x^{c,\ell,i} = (\underline{x}_1^\ell, \dots, \underline{x}_{i-1}^\ell, \xi_i^\ell, \underline{x}_{i+1}^\ell, \dots, \underline{x}_n^\ell) \text{ satisfies } N(x^{c,\ell,i}) > 0.$$

5:     if such a point is found then
6:        $\mathcal{C}^+ \leftarrow \mathcal{C}^+ \cup \{x^{c,\ell,i}\}$ 
7:     else
8:        $\xi_i^\ell \leftarrow \bar{x}_i^\ell$ 
9:     end if
10:  end for
11:  Define  $\hat{x}^\ell = (\xi_1^\ell, \dots, \xi_n^\ell)$ 
12:  if  $\hat{x}^\ell = \bar{x}^\ell$  then
13:    return FAIL
14:  end if
15:   $\mathcal{H}^+ \leftarrow \mathcal{H}^+ \cup \{[\underline{x}^\ell, \hat{x}^\ell]\}$ 
16: end for
17: return  $\mathcal{H}^+, \mathcal{C}^+$ 

```

J.4 Arbitrary Hyper-Rectangular Gridding

Algorithm 4 Arbitrary refinement of state space grid

Inputs: The current unresolved hyper-rectangular collection $\mathcal{H} = \{[\underline{x}^\ell, \bar{x}^\ell]\}_{\ell \in \mathcal{P} \setminus \mathcal{U}}$; the current shrinking-constraint set $\mathcal{C}$; and constants $\epsilon > 0$ and $C_{\max} > 1$ defining the random split range.

Output: A refined unresolved collection $\mathcal{H}^+$ and an unchanged shrinking-constraint set $\mathcal{C}^+$.

```

1: Initialize  $\mathcal{H}^+ \leftarrow \emptyset$  and  $\mathcal{C}^+ \leftarrow \mathcal{C}$ 
2: for each box  $[\underline{x}^\ell, \bar{x}^\ell] \in \mathcal{H}$  do
3:   Let  $d = \arg \max_k (\bar{x}_k^\ell - \underline{x}_k^\ell)$ 
4:   Sample  $r_x \sim \text{Uniform}(1 + \epsilon, C_{\max})$ 
5:   Compute  $c_d^\ell = \underline{x}_d^\ell + \frac{\bar{x}_d^\ell - \underline{x}_d^\ell}{r_x}$ 
6:   Let  $\bar{x}^{\ell,1} = \bar{x}^\ell$  with  $\bar{x}_d^{\ell,1} = c_d^\ell$ 
7:   Let  $\underline{x}^{\ell,2} = \underline{x}^\ell$  with  $\underline{x}_d^{\ell,2} = c_d^\ell$ 
8:   Define  $H_1^\ell = [\underline{x}^\ell, \bar{x}^{\ell,1}]$ ,  $H_2^\ell = [\underline{x}^{\ell,2}, \bar{x}^\ell]$ .
9:    $\mathcal{H}^+ \leftarrow \mathcal{H}^+ \cup \{H_1^\ell, H_2^\ell\}$ 
10: end for
11: return  $\mathcal{H}^+, \mathcal{C}^+$ 

```

Algorithm 5 Arbitrary refinement of input space grid

Inputs: The current input cover $\mathcal{J}$; and constants $\epsilon > 0$ and $C_{\max} > 1$ defining the random split range.

Output: A refined input cover $\mathcal{J}^+$.

```

1: Initialize  $\mathcal{J}^+ \leftarrow \emptyset$ 
2: for each box  $[\underline{u}^q, \bar{u}^q] \in \mathcal{J}$  do
3:   Let  $d = \arg \max_k (\bar{u}_k^q - \underline{u}_k^q)$ 
4:   Sample  $r_u \sim \text{Uniform}(1 + \epsilon, C_{\max})$ 
5:   Compute  $c_d^q = \underline{u}_d^q + \frac{\bar{u}_d^q - \underline{u}_d^q}{r_u}$ 
6:   Let  $\bar{u}^{q,1} = \bar{u}^q$  with  $\bar{u}_d^{q,1} = c_d^q$ 
7:   Let  $\underline{u}^{q,2} = \underline{u}^q$  with  $\underline{u}_d^{q,2} = c_d^q$ 
8:   Define  $J_1^q = [\underline{u}^q, \bar{u}^{q,1}]$ ,  $J_2^q = [\underline{u}^{q,2}, \bar{u}^q]$ .
9:    $\mathcal{J}^+ \leftarrow \mathcal{J}^+ \cup \{J_1^q, J_2^q\}$ 
10: end for
11: return  $\mathcal{J}^+$ 

```

K The role of the inductive parameter K in sample complexity

In this section, we investigate the effect of the inductive parameter k on the sample complexity. To this end, we consider the 2-D traffic system described in Appendix G. As shown previously, our method verifies safety for this system using an extremely coarse rectangular gridding (two rectangles covering the unsafe region, one rectangle for the initial set, and one rectangle for the remaining safe region), enabled by the long inductive parameter $k = 100$. In contrast, for the same coarse gridding, ordinary one-step barrier conditions (based only on extreme points and a single-step horizon) are insufficient to certify safety. Motivated by this limitation, we now refine the gridding over the state space using Algorithm 4 and try to learn a classic neural barrier certificate from the resulting denser set of samples. Figure 4 shows that, for a classic (one-step) barrier certificate (i.e., the inductive parameter $K = 1$), safety verification requires a very dense rectangular gridding (103,000 boxes), which leads to a sample-intensive procedure and becomes impractical for high-dimensional systems. In this experiment, we use a simplified Max-Monotone Network based on the direct max-input branch in Definition 4.1, with two hidden layers of widths 200 and 100, respectively, and Leaky ReLU activations ($\alpha = 0.1$).

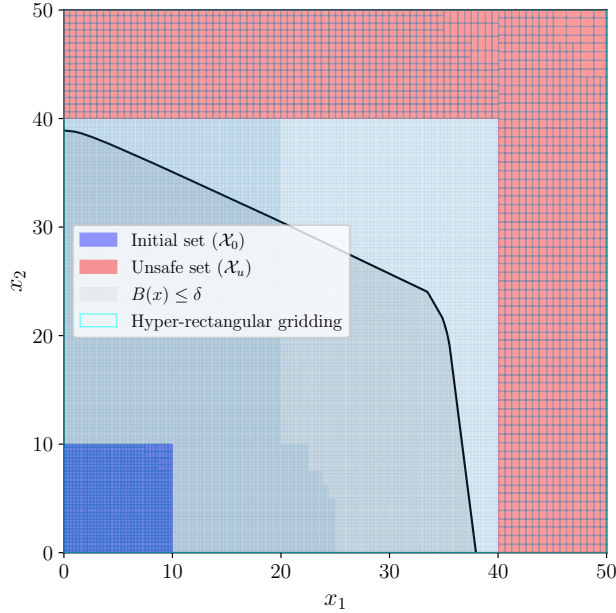


Figure 4: Sub-level set of the learned neural barrier certificate (one-step barrier) over the state space. The parameters $\gamma = 5$, $\delta = 9.5$, and $\eta = 10$ are used in this experiment.

As illustrated in Figure 4, without the benefit of an inductive barrier (even though we still exploit monotonicity of both the system and the barrier function), the required sample complexity remains high, even for this simple 2-D case study. This highlights the crucial role of the inductive parameter K in mitigating sample complexity.

L Proof of Theorem 4.3

The proof of Theorem 4.3 is provided below.

Proof. We divide the proof into two parts.

(i) *Correctness.* Suppose Algorithm 1, using Algorithm 3 as the refinement routine, returns $(N, \kappa, L_{\text{total}})$. By the termination condition of the algorithm, $L_{\text{total}} = L_1 + L_2 + L_3 + L_{\text{shrink}} = 0$. Since all loss terms are nonnegative, it follows that

$$L_1 = L_2 = L_3 = L_{\text{shrink}} = 0.$$

By the standing result already established in the paper, $L_1 = L_2 = L_3 = 0$ implies that (N, κ) satisfies the K -inductive control barrier conditions (6)–(8). Moreover, $L_{\text{shrink}} = 0$ implies

$$N(x_{(q)}^{c,\ell,i}) \geq \eta > 0 \quad \forall x_{(q)}^{c,\ell,i} \in \mathcal{C},$$

so all shrinking constraints generated in previous iterations are preserved and no previously certified shrinking step is invalidated by retraining. Therefore, the returned pair (N, κ) satisfies the required control barrier conditions together with all accumulated shrinking constraints. The safety of Σ with respect to $\mathcal{X}_u$ then follows from the control barrier theorem.

(ii) *Linear simulator-query complexity.* Let $M := |\mathcal{P}_0|$. By construction, each unresolved box produces exactly one shrunken unresolved box at the next shrinking step. Hence, after each call to Algorithm 3, the number of unresolved boxes remains equal to M , which is independent of n .

Now fix one call to Algorithm 3. For each unresolved box and each coordinate $i \in \{1, \dots, n\}$, the algorithm performs one one-dimensional search for $\xi_i^{\ell,(p)}$. Since the search is one-dimensional and the algorithmic budgets are fixed independently of n , each such search requires a constant number of simulator queries with respect to n . Therefore, one call to Algorithm 3 requires at most cMn simulator queries for some constant $c > 0$ independent of n .

Finally, the number of calls to Algorithm 3, inductive-parameter updates, and training epochs are bounded by the fixed budgets $n_{\max}^{\mathcal{P}}$, $K_{\max}$, and N_{ep} , respectively. Since these quantities and the cardinalities of $\mathcal{I}, \mathcal{U}, \mathcal{P}_0$ are all independent of n , they only contribute multiplicative constants. Hence, the total number of simulator queries is bounded by a constant multiple of n , and is therefore $\mathcal{O}(n)$. $\square$

M Correctness and Linear Simulator-Query Complexity for Verification

Theorem M.1 (Correctness and linear simulator-query complexity of Algorithm 2). *Consider a monotone discrete-time system $\Sigma = (\mathcal{X}, \mathcal{X}_0, U, f)$, $\mathcal{X} \subseteq \mathbb{R}^n$, and suppose that*

$$\mathcal{X}_0 \subseteq \bigcup_{i \in \mathcal{I}} [\underline{x}_0^i, \bar{x}_0^i], \quad \mathcal{X}_u \subseteq \bigcup_{j \in \mathcal{U}} [\underline{x}_u^j, \bar{x}_u^j], \quad \mathcal{X} \setminus \mathcal{X}_u \subseteq \bigcup_{\ell \in \mathcal{P}_0} [\underline{x}^\ell, \bar{x}^\ell],$$

and

$$U \subseteq \bigcup_{q \in \mathcal{Q}_0} [\underline{u}^q, \bar{u}^q],$$

for finite index sets $\mathcal{I}, \mathcal{U}, \mathcal{P}_0$, and $\mathcal{Q}_0$ whose cardinalities are independent of n . Then:

- (i) *If Algorithm 2, using either Algorithm 3 or Algorithm 4 as the state-refinement routine and Algorithm 5 as the input-refinement routine, returns (N, L_{total}^v) , then the system Σ is robustly safe with respect to $\mathcal{X}_u$.*
- (ii) *The total number of simulator queries performed by Algorithm 2, using Algorithm 3 as the state-refinement routine and Algorithm 5 as the input-refinement routine, is $\mathcal{O}(n)$.*

Proof. We divide the proof into two parts.

(i) *Correctness.* Suppose Algorithm 2, using either Algorithm 3 or Algorithm 4 as the state-refinement routine and Algorithm 5 as the input-refinement routine, returns (N, L_{total}^v) . By the termination condition of the algorithm,

$$L_{\text{total}}^v = L_1^v + L_2^v + L_3^v + L_{\text{shrink}} = 0.$$

Since all loss terms are nonnegative, it follows that

$$L_1^v = L_2^v = L_3^v = L_{\text{shrink}} = 0.$$

By Theorem I.1, $L_1^v = L_2^v = L_3^v = 0$ implies that N satisfies the K -inductive robust barrier conditions (16)–(18). Moreover, $L_{\text{shrink}} = 0$ implies

$$N(x_{(q)}^{c,\ell,i}) \geq \eta > 0, \quad \forall x_{(q)}^{c,\ell,i} \in \mathcal{C},$$

so all accumulated shrinking constraints, if any, are preserved throughout the training process. Therefore, the returned certificate N satisfies the required robust barrier conditions together with all accumulated shrinking constraints. The robust safety of Σ with respect to $\mathcal{X}_u$ then follows from the robust barrier theorem.

(ii) *Linear simulator-query complexity under monotone shrinking refinement.* Let $M := |\mathcal{P}_0|$. When Algorithm 3 is used as the state-refinement routine, each unresolved box produces exactly one refined unresolved box at the next refinement step. Hence, after each call to Algorithm 3, the number of unresolved boxes remains equal to M , which is independent of n .

Now fix one call to Algorithm 3. For each unresolved box and each coordinate $i \in \{1, \dots, n\}$, the algorithm performs one one-dimensional search for $\xi_i^{\ell, (p)}$. Since the search is one-dimensional and the algorithmic budgets are fixed independently of n , each such search requires a constant number of simulator queries with respect to n . Therefore, one call to Algorithm 3 requires at most $c_1 M n$ simulator queries for some constant $c_1 > 0$ independent of n .

In the robust verification losses, the current input cover $\mathcal{J}$ also appears. Since the input-refinement budget $n_{\max}^{\mathcal{Q}}$ is fixed independently of n , the cardinality of $\mathcal{J}$ remains bounded independently of n . Therefore, evaluating the loss terms over the input cover contributes only a constant multiplicative factor.

Finally, the numbers of refinement iterations, inductive-parameter updates, and training epochs are bounded by the fixed budgets $n_{\max}^{\mathcal{P}}$, $n_{\max}^{\mathcal{Q}}$, $K_{\max}$, and N_{ep} , respectively. Since these quantities and the cardinalities of $\mathcal{I}$, $\mathcal{U}$, $\mathcal{P}_0$, and $\mathcal{Q}_0$ are all independent of n , they contribute only multiplicative constants. Hence, the total number of simulator queries is bounded by a constant multiple of n , and is therefore $\mathcal{O}(n)$. $\square$

N Numerical Experiment Details

In this appendix, we provide details of our numerical experiments in Section 5.

N.1 Baseline methods and comparisons

The baseline methods considered for comparison are Method 1, the counterexample-guided approach [25], and Method 2, the Lipschitz-based method [7]. Although [25] requires explicit access to the dynamics of the system during its formal guarantee stage, we include it for comparison since its barrier construction is still data-driven. In particular, [25] relies on fixed random sampling and SMT-based verification, which fails to provide guarantees as the dimension grows.

All experiments of the proposed method were conducted on a MacBook Pro equipped with an Apple M3 Max processor and 48 GB RAM. All experiments for the method of [25] were conducted on a workstation running Ubuntu 20.04, equipped with Intel Xeon Gold 6238R CPUs, two NVIDIA Quadro RTX 8000 GPUs, and 160 GB of system memory.

In all experiments, the barrier certificate $N : \mathbb{R}^n \rightarrow \mathbb{R}$, and in the safe synthesis setting also the feedback controller $\kappa : \mathbb{R}^n \rightarrow \mathbb{R}^n$, are parameterized by the Max-Monotone Network architecture with three branches: one direct max-input branch and two affine max branches with projection dimensions $m_1 = 0.2n$ and $m_2 = 0.1n$, except for the Lotka–Volterra experiments reported in Table 1, where we use the fixed dimensions $m_1 = 10$ and $m_2 = 20$. Each branch is followed by a monotone subnetwork with two hidden layers of 200 neurons and Leaky ReLU activations ($\alpha = 0.1$).

N.2 Freeway Traffic Flow Model

Model. We consider the freeway traffic network $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ illustrated in Fig. 5. The state is $q(t) = [q_1(t), \dots, q_n(t)]^\top \in \mathbb{R}^n$, where $q_i(t) \geq 0$ denotes the vehicle density in segment i . The state space is $\mathcal{X} = [0, 50]^n$.

The dynamics $f : \mathcal{X} \times U \rightarrow \mathcal{X}$ are given by

$$q_1(t+1) = q_1(t) + \tau(F_{0 \rightarrow 1} - F_{1 \rightarrow 2}(q(t))), \quad (46)$$

$$q_i(t+1) = q_i(t) + \tau(F_{i-1 \rightarrow i}(q(t)) - F_{i \rightarrow i+1}(q(t))), \quad i = 2, \dots, n-1, \quad (47)$$

$$q_n(t+1) = q_n(t) + \tau(F_{n-1 \rightarrow n}(q(t)) - F_{n \rightarrow 0}(q(t))). \quad (48)$$

The input corresponds to the on-ramp inflow $F_{0 \rightarrow 1} \in U = [0, 7.5]$, with sampling time $\tau = 0.5$.

The inter-segment flow is defined by

$$F_{i \rightarrow i+1}(q) = \min\{\varphi(q_i), \sigma(q_{i+1})\}, \quad (49)$$

$$F_{n \rightarrow 0}(q) = \varphi(q_n), \quad (50)$$

where

$$\varphi(x) = 15(1 - e^{-0.07x}), \quad \sigma(x) = \max\left\{0, 15\left(1 - \left(\frac{x}{50}\right)^3\right)\right\}.$$

The initial set is $\mathcal{X}_0 = [0, 5]^n$ and the unsafe set is $\mathcal{X}_u = \mathcal{X} \setminus [0, 40]^n$.

Monotonicity. The system is monotone with respect to the standard partial order; see [13] for a detailed proof.

N.3 Kuramoto Oscillator Networks

Fully connected network. We consider a discrete-time Kuramoto oscillator network $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, U, f)$ consisting of n oscillators with fully connected coupling [51]. Let $\theta(t) = [\theta_1(t); \dots; \theta_n(t)] \in \mathbb{R}^n$ denote the vector of oscillator phases at time t , where $\theta_i(t) \in [0, \frac{\pi}{4}]$.

The state, initial, and input sets are

$$\mathcal{X} = [0, \frac{\pi}{4}]^n, \quad \mathcal{X}_0 = [0, \frac{\pi}{16}]^n, \quad U = [-\frac{\pi}{4}, \frac{\pi}{4}]^n.$$

The dynamics are given by

$$\theta_i(t+1) = \theta_i(t) + \frac{\tau K_c}{n} \sum_{j \neq i} \sin(\theta_j(t) - \theta_i(t)) + \tau u_i(t),$$

where $\tau > 0$ is the sampling time and $K_c > 0$ is the coupling strength.

Monotonicity. For $j \neq i$, we have

$$\frac{\partial f_i}{\partial \theta_j} = \frac{\tau K_c}{n} \cos(\theta_j - \theta_i) \geq 0,$$

since $\theta_j - \theta_i \in [-\frac{\pi}{4}, \frac{\pi}{4}]$.

For the diagonal term,

$$\frac{\partial f_i}{\partial \theta_i} = 1 - \frac{\tau K_c}{n} \sum_{j \neq i} \cos(\theta_j - \theta_i) \geq 1 - \tau K_c.$$

Thus, if $\tau K_c \leq 1$, all entries of the Jacobian are nonnegative, and the system is monotone. Moreover, $\frac{\partial f_i}{\partial u_i} = \tau > 0$, so the system is also monotone with respect to the input.

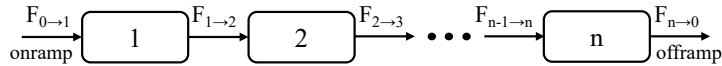


Figure 5: A line digraph illustrating an n -dimensional traffic network.

Networks with structured coupling (IEEE testbeds). We also consider Kuramoto networks with non-uniform coupling structures derived from IEEE power-system testbeds. In this case, the dynamics are given by

$$\theta_i(t+1) = \theta_i(t) + \tau u_i(t) + \frac{\tau K_c}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} a_{ij} \sin(\theta_j(t) - \theta_i(t)),$$

where $\mathcal{N}_i$ denotes the set of neighbors of node i and $a_{ij} \in \{0, 1\}$ indicates connectivity.

The state, initial, input, and unsafe sets are identical to the fully connected case.

Monotonicity. For $j \neq i$,

$$\frac{\partial f_i}{\partial \theta_j} = \frac{\tau K_c}{|\mathcal{N}_i|} a_{ij} \cos(\theta_j - \theta_i) \geq 0.$$

For the diagonal term,

$$\frac{\partial f_i}{\partial \theta_i} = 1 - \frac{\tau K_c}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} a_{ij} \cos(\theta_j - \theta_i) \geq 1 - \tau K_c.$$

Thus, under the same condition $\tau K_c \leq 1$, the system is monotone. As before, $\frac{\partial f_i}{\partial u_i} = \tau > 0$, so monotonicity with respect to the input also holds.

N.4 Lotka–Volterra Population Dynamics

Model. We consider the n -dimensional Lotka–Volterra population model in discrete time, modeled as the system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$ with state $x(t) \in \mathbb{R}_{\geq 0}^n$. Let $A = [a_{ij}] \in \mathbb{R}^{n \times n}$ denote the interaction matrix and let $r \in \mathbb{R}^n$ be the intrinsic growth-rate vector. The discrete-time dynamics are given by

$$x(t+1) = f(x(t)) := x(t) + \tau [x(t)](Ax(t) + r),$$

where $\tau > 0$ is the sampling time and $[x(t)] := \text{diag}(x_1(t), \dots, x_n(t))$. Equivalently, for each $i \in \{1, \dots, n\}$,

$$x_i(t+1) = x_i(t) + \tau x_i(t) \left(\sum_{j=1}^n a_{ij} x_j(t) + r_i \right).$$

Monotonicity. A sufficient condition for f to be monotone (with respect to the standard partial order) on a set $\mathcal{X}$ is

$$Df(x) \geq 0, \quad \forall x \in \mathcal{X},$$

where the inequality is understood elementwise.

The Jacobian of f is given by

$$Df(x) = I + \tau \left([x]A + \text{diag}(Ax + r) \right).$$

For $i \neq j$, the off-diagonal entries are

$$\frac{\partial f_i}{\partial x_j}(x) = \tau x_i a_{ij} \geq 0,$$

which holds for all $x \geq 0$ whenever A is Metzler, i.e., $a_{ij} \geq 0$ for all $i \neq j$.

For the diagonal terms,

$$\frac{\partial f_i}{\partial x_i}(x) = 1 + \tau \left(x_i a_{ii} + (Ax + r)_i \right),$$

and thus we require

$$1 + \tau \left(x_i a_{ii} + (Ax + r)_i \right) \geq 0, \quad \forall x \in \mathcal{X}.$$

Now consider the hyper-rectangle $\mathcal{X} = [0, \bar{x}]^n$. Since A is Metzler and $x \geq 0$, we have

$$(Ax + r)_i = r_i + a_{ii}x_i + \sum_{j \neq i} a_{ij}x_j \geq r_i + a_{ii}x_i,$$

which implies

$$x_i a_{ii} + (Ax + r)_i \geq r_i + 2a_{ii}x_i.$$

If $a_{ii} \leq 0$, we further obtain

$$r_i + 2a_{ii}x_i \geq r_i + 2a_{ii}\bar{x}.$$

Therefore, a sufficient condition for monotonicity on $\mathcal{X} = [0, \bar{x}]^n$ is

$$1 + \tau(r_i + 2a_{ii}\bar{x}) \geq 0, \quad \forall i \in \{1, \dots, n\}.$$

Equivalently, it suffices to choose τ such that

$$\tau \leq \min_{i: r_i + 2a_{ii}\bar{x} < 0} \frac{1}{-(r_i + 2a_{ii}\bar{x})},$$

with the convention that the bound is $+\infty$ if $r_i + 2a_{ii}\bar{x} \geq 0$ for all i .

N.5 Ribosome Flow Model

Model. We consider a Ribosome Flow Model (RFM) $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$ consisting of n sites. Let $x(t) = [x_1(t); \dots; x_n(t)]$ denote the state at time step t , where $x_i(t) \in [0, 1]$ represents the occupancy level of site i . The dynamics are given by

$$\begin{aligned} x_1(t+1) &= x_1(t) + h(\lambda_0(1 - x_1(t)) - \lambda_1 x_1(t)(1 - x_2(t))), \\ x_i(t+1) &= x_i(t) + h(\lambda_{i-1}x_{i-1}(t)(1 - x_i(t)) - \lambda_i x_i(t)(1 - x_{i+1}(t))), \quad i = 2, \dots, n-1, \\ x_n(t+1) &= x_n(t) + h(\lambda_{n-1}x_{n-1}(t)(1 - x_n(t)) - \lambda_n x_n(t)). \end{aligned}$$

where $\lambda_i > 0$ are transition rates and $h > 0$ is the sampling time.

In our experiments, we sample λ_i independently from the uniform distribution $\lambda_i \sim \mathcal{U}(0.8, 1.2)$. The step size h is chosen to satisfy the monotonicity condition

$$h \leq \left(\max_i (\lambda_i + \lambda_{i+1}) \right)^{-1}.$$

We consider the state space $\mathcal{X} = [0, 1]^n$ and the initial set $\mathcal{X}_0 = [0, 0.2]^n$. The unsafe set is defined as

$$\mathcal{X}_u = [0, 1]^n \setminus [0, 0.9]^n.$$

Monotonicity. The monotonicity of the system follows from the Jacobian of the evaluation map. For $j \neq i$, we have

$$\frac{\partial f_i}{\partial x_j}(x) \geq 0,$$

since all off-diagonal terms are of the form $h\lambda_i x_i$ or $h\lambda_i(1 - x_i)$, which are nonnegative on $[0, 1]^n$.

For the diagonal terms,

$$\frac{\partial f_i}{\partial x_i}(x) = 1 - h\lambda_{i-1}x_{i-1} - h\lambda_i(1 - x_{i+1}),$$

and since $x_i \in [0, 1]$, it follows that

$$\frac{\partial f_i}{\partial x_i}(x) \geq 1 - h(\lambda_{i-1} + \lambda_i).$$

Thus, under the condition

$$h \leq (\lambda_{i-1} + \lambda_i)^{-1}, \quad \forall i,$$

all entries of the Jacobian are nonnegative, and the system is monotone on $\mathcal{X}$.

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